

STREAMING VIDEO SWITCHER  
**V-160HD** Version 3.5 and later

DIRECT STREAMING VIDEO SWITCHER  
**V-80HD** Version 1.3 and later

DIRECT STREAMING AV MIXER  
**VR-120HD** Version 2.2 and later

DIRECT STREAMING AV MIXER  
**VR-6HD** Version 2.2 and later

VIDEO SWITCHER  
**V-1-4K** Version 1.0 and later

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# Basic control commands

The V-160HD/V-80HD/VR-120HD/VR-6HD/V-1-4K supports two types of remote interface communication: LAN and RS-232.

By using the LAN CONTROL port or the RS-232 connector and sending the specified command from the controlling device to the V-160HD/V-80HD/VR-120HD/VR-6HD/V-1-4K, you can remotely control these products.

- \* This manual describes the basic control commands for the products in question.
- \* You can use the basic control commands at the same time as the regular commands.

## MEMO

Refer to the remote control guide for each product for how to configure the LAN or RS-232 interface.

## Command format

Commands are formatted using the configuration shown below.

- \* Commands are all in ASCII code.
- \* The commands are common to both the LAN and RS-232 interfaces.

**Command code:parameter;parameter;**

### Command code

Specifies the command type (single-byte alphanumeric characters).

### Parameter

This is appended to a command that requires a parameter.

The command and the parameter portion are separated by a colon.

When there are multiple parameters, they are separated by commas.

### “;” (semicolon)

The semicolon is the code that indicates the end of the command.

For the response commands sent from the product in question, a carriage return code (0x0A, LF, ¥n) is sent after the semicolon.

## VIDEO

| QVISRC Get Video Input Source |                       |          |                                                      |
|-------------------------------|-----------------------|----------|------------------------------------------------------|
| Transmitted command           | <b>QVISRC:a;</b>      |          |                                                      |
|                               | <b>a</b>              | V-160HD  | 0–51                                                 |
|                               |                       | V-80HD   | 0–56                                                 |
|                               |                       | VR-120HD | 0–36                                                 |
|                               |                       | VR-6HD   | 0–28                                                 |
|                               |                       | V-1-4K   | 0–15                                                 |
| Response command              | <b>VISRC:a,b;ACK;</b> |          |                                                      |
|                               | <b>a</b>              | V-160HD  | 0–51                                                 |
|                               |                       | V-80HD   | 0–56                                                 |
|                               |                       | VR-120HD | 0–36                                                 |
|                               |                       | VR-6HD   | 0–28                                                 |
|                               |                       | V-1-4K   | 0–14                                                 |
|                               | <b>b</b>              | V-160HD  | HDMI1–8, SDI1–8, STILL1–16, INPUT1–20                |
|                               |                       | V-80HD   | HDMI1–4, SDI1–4, STILL1–32, V.PLAYER, SRT, INPUT1–16 |
|                               |                       | VR-120HD | HDMI1–6, SDI1–6, STILL1–16, V.PLAYER, SRT, INPUT1–8  |
|                               |                       | VR-6HD   | HDMI1–6, STILL1–16, V.PLAYER, SRT, INPUT1–6          |
|                               |                       | V-1-4K   | HDMI1–5, STILL1–2, INPUT1–8, BLACK                   |

## Examples of settings

|                     |                           |                                        |
|---------------------|---------------------------|----------------------------------------|
| Transmitted command | <b>QVISRC:0;</b>          | Checks the zeroth Video Input Source   |
| Response command    | <b>VISRC:0,HDMI1;ACK;</b> | The zeroth Video Input Source is HDMI1 |

| QVIST Get Video Input Status |                      |                                 |                 |
|------------------------------|----------------------|---------------------------------|-----------------|
| Transmitted command          | <b>QVIST:a;</b>      |                                 |                 |
|                              | <b>a</b>             | V-160HD                         | HDMI1–8, SDI1–8 |
|                              |                      | V-80HD                          | HDMI1–4, SDI1–4 |
|                              |                      | VR-120HD                        | HDMI1–6, SDI1–6 |
|                              |                      | VR-6HD                          | HDMI1–6         |
|                              |                      | V-1-4K                          | HDMI1–5         |
| Response command             | <b>VIST:a,b;ACK;</b> |                                 |                 |
|                              | <b>a</b>             | V-160HD                         | HDMI1–8, SDI1–8 |
|                              |                      | V-80HD                          | HDMI1–4, SDI1–4 |
|                              |                      | VR-120HD                        | HDMI1–6, SDI1–6 |
|                              |                      | VR-6HD                          | HDMI1–6         |
|                              |                      | V-1-4K                          | HDMI1–5         |
|                              | <b>b</b>             | NOSIGNAL, DETECTED, UNSUPPORTED |                 |

## Examples of settings

|                     |                                 |                                          |
|---------------------|---------------------------------|------------------------------------------|
| Transmitted command | <b>QVIST:HDMI1;</b>             | Check the HDMI1 Video Input Status       |
| Response command    | <b>VIST:HDMI1,DETECTED;ACK;</b> | The HDMI1 Video Input Status is DETECTED |

## Basic control commands

| VFL                 |        | Set Video Fader Level |        |
|---------------------|--------|-----------------------|--------|
| Transmitted command | VFL:a; |                       |        |
|                     | a      | V-160HD               |        |
|                     |        | V-80HD                |        |
|                     |        | VR-120HD              | 0–2047 |
|                     |        | VR-6HD                |        |
|                     | V-1-4K | 0–4095                |        |
| Response command    | ACK;   |                       |        |

### Examples of settings

|                     |                  |                                 |
|---------------------|------------------|---------------------------------|
| Transmitted command | <b>VFL:1024;</b> | Set Video Fader Level to “1024” |
| Response command    | <b>ACK;</b>      |                                 |

| QVFLGet Video Fader Level |            |          |        |
|---------------------------|------------|----------|--------|
| Transmitted command       | QVFL;      |          |        |
| Response command          | VFL:a;ACK; |          |        |
|                           | a          | V-160HD  |        |
|                           |            | V-80HD   |        |
|                           |            | VR-120HD | 0–2047 |
|                           |            | VR-6HD   |        |
|                           | V-1-4K     | 0–4095   |        |

### Examples of settings

|                     |                      |                             |
|---------------------|----------------------|-----------------------------|
| Transmitted command | <b>QVFL;</b>         | Check Video Fader Level     |
| Response command    | <b>VFL:1024;ACK;</b> | Video Fader Level is “1024” |

| PGM Select PGM video channel |        |          |                                                      |
|------------------------------|--------|----------|------------------------------------------------------|
| Transmitted command          | PGM:a; |          |                                                      |
|                              | a      | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                              |        | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                              |        | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                              |        | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |
|                              |        | V-1-4K   | HDMI1-5, STILL1-2, INPUT1-8                          |
| Response command             | ACK;   |          |                                                      |

### Examples of settings

|                     |                    |                                   |
|---------------------|--------------------|-----------------------------------|
| Transmitted command | <b>PGM:INPUT2;</b> | Select “INPUT2” for the PGM video |
| Response command    | <b>ACK;</b>        |                                   |

| QPGM; Get channel of the PGM video |              |
|------------------------------------|--------------|
| Transmitted command                | <b>QPGM;</b> |

|                  |                     |          |                                           |
|------------------|---------------------|----------|-------------------------------------------|
| Response command | <b>PGM:a,b;ACK;</b> |          |                                           |
|                  | <b>a</b>            | V-160HD  | HDMI1-8, SDI1-8, STILL1-16                |
|                  |                     | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT |
|                  |                     | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT |
|                  |                     | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT         |
|                  |                     | V-1-4K   | HDMI1-5, STILL1-2                         |
|                  | <b>b</b>            | V-160HD  | INPUT1-20                                 |
|                  |                     | V-80HD   | INPUT1-16                                 |
|                  |                     | VR-120HD | INPUT1-8                                  |
|                  |                     | VR-6HD   | INPUT1-6                                  |
|                  |                     | V-1-4K   | INPUT1-8                                  |

\* Only when selected as an input channel

## Examples of settings

|                     |                              |                                     |
|---------------------|------------------------------|-------------------------------------|
| Transmitted command | <b>QPGM;</b>                 | Check channel of PGM video          |
| Response command    | <b>PGM:HDMI2,INPUT2;ACK;</b> | PGM video channel is INPUT2 (HDMI2) |

| PST                 |        | Select PST video channel |                                                      |
|---------------------|--------|--------------------------|------------------------------------------------------|
| Transmitted command | PST:a; |                          |                                                      |
|                     | a      | V-160HD                  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                     |        | V-80HD                   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                     |        | VR-120HD                 | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                     |        | VR-6HD                   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |
|                     |        | V-1-4K                   | HDMI1-5, STILL1-2, INPUT1-8                          |
| Response command    | ACK;   |                          |                                                      |

| QPST                                     |              | Get PST video channel |                                           |
|------------------------------------------|--------------|-----------------------|-------------------------------------------|
| Transmitted command                      | QPST;        |                       |                                           |
| Response command                         | PST:a,b;ACK; |                       |                                           |
|                                          | a            | V-160HD               | HDMI1-8, SDI1-8, STILL1-16                |
|                                          |              | V-80HD                | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT |
|                                          |              | VR-120HD              | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT |
|                                          |              | VR-6HD                | HDMI1-6, STILL1-16, V.PLAYER, SRT         |
|                                          |              | V-1-4K                | HDMI1-5, STILL1-2                         |
|                                          | b            | V-160HD               | INPUT1-20                                 |
|                                          |              | V-80HD                | INPUT1-16                                 |
|                                          |              | VR-120HD              | INPUT1-8                                  |
|                                          |              | VR-6HD                | INPUT1-6                                  |
|                                          |              | V-1-4K                | INPUT1-8                                  |
| * Only when selected as an input channel |              |                       |                                           |

\* Only when selected as an input channel

## Basic control commands

| AUX Select channel of video to transmit to AUX bus (V-160HD/V-80HD/VR-120HD/VR-6HD) |                 |          |                                                               |
|-------------------------------------------------------------------------------------|-----------------|----------|---------------------------------------------------------------|
| Transmitted command                                                                 | <b>AUX:a,b;</b> |          |                                                               |
|                                                                                     | <b>a</b>        | V-160HD  | AUX1-3                                                        |
|                                                                                     |                 | V-80HD   | AUX1-2                                                        |
|                                                                                     |                 | VR-120HD | AUX1-3                                                        |
|                                                                                     |                 | VR-6HD   | AUX1                                                          |
|                                                                                     | <b>b</b>        | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20, PGMLINK                |
|                                                                                     |                 | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16, PGMLINK |
|                                                                                     |                 | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8, PGMLINK  |
|                                                                                     |                 | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6, PGMLINK          |
| Response command                                                                    | <b>ACK;</b>     |          |                                                               |

\* PGMLINK can be set when the AUX Linked PGM parameter is enabled.

| QAUX Get channel of video to transmit to AUX bus (V-160HD/V-80HD/VR-120HD/VR-6HD) |                       |          |                                           |
|-----------------------------------------------------------------------------------|-----------------------|----------|-------------------------------------------|
| Transmitted command                                                               | <b>QAUX:a;</b>        |          |                                           |
|                                                                                   | <b>a</b>              | V-160HD  | AUX1-3                                    |
|                                                                                   |                       | V-80HD   | AUX1-2                                    |
|                                                                                   |                       | VR-120HD | AUX1-3                                    |
|                                                                                   |                       | VR-6HD   | AUX1                                      |
| Response command                                                                  | <b>AUX:a,b,c;ACK;</b> |          |                                           |
|                                                                                   | <b>a</b>              | V-160HD  | AUX1-3                                    |
|                                                                                   |                       | V-80HD   | AUX1-2                                    |
|                                                                                   |                       | VR-120HD | AUX1-3                                    |
|                                                                                   |                       | VR-6HD   | AUX1                                      |
|                                                                                   | <b>b</b>              | V-160HD  | HDMI1-8, SDI1-8, STILL1-16                |
|                                                                                   |                       | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT |
|                                                                                   |                       | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT |
|                                                                                   |                       | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT         |
|                                                                                   | <b>c</b>              | V-160HD  | INPUT1-20                                 |
|                                                                                   |                       | V-80HD   | INPUT1-16                                 |
|                                                                                   |                       | VR-120HD | INPUT1-8                                  |
|                                                                                   |                       | VR-6HD   | INPUT1-6                                  |

\* Only when selected as an input channel

| ATO Auto-switch video (enabled when SPLIT is OFF) |             |
|---------------------------------------------------|-------------|
| Transmitted command                               | <b>ATO;</b> |
| Response command                                  | <b>ACK;</b> |

| ATO Auto-switch video (enabled when SPLIT is OFF) |               |          |                                                      |
|---------------------------------------------------|---------------|----------|------------------------------------------------------|
| Transmitted command                               | <b>ATO:a;</b> |          |                                                      |
|                                                   | <b>a</b>      | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                                                   |               | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                                                   |               | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                                                   |               | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |
|                                                   |               | V-1-4K   | HDMI1-5, STILL1-2, INPUT1-8                          |
| Response command                                  | <b>ACK;</b>   |          |                                                      |

| ATO                 |          |          |                                                      |  | Auto-switch video (enabled when SPLIT is OFF) |                                                                 |  |  |  |
|---------------------|----------|----------|------------------------------------------------------|--|-----------------------------------------------|-----------------------------------------------------------------|--|--|--|
| Transmitted command | ATO:a,b; |          |                                                      |  |                                               |                                                                 |  |  |  |
|                     | a        | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |  |                                               |                                                                 |  |  |  |
|                     |          | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |  |                                               |                                                                 |  |  |  |
|                     |          | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |  |                                               |                                                                 |  |  |  |
|                     |          | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |  |                                               |                                                                 |  |  |  |
|                     |          | V-1-4K   | HDMI1-5, STILL1-2, INPUT1-8                          |  |                                               |                                                                 |  |  |  |
|                     | b        |          |                                                      |  | -1, 0-40                                      | -1: the setting for Transition Time is used<br>0-40: 0.0-4.0sec |  |  |  |
| Response command    | ACK;     |          |                                                      |  |                                               |                                                                 |  |  |  |

## Examples of settings

|                     |                       |                                                     |
|---------------------|-----------------------|-----------------------------------------------------|
| Transmitted command | <b>ATO:INPUT1,10;</b> | Auto-switch between videos for INPUT1 every 1.0 sec |
| Response command    | <b>ACK;</b>           |                                                     |

| CUT Switch between video using CUT (enabled when SPLIT is OFF) |             |  |  |  |
|----------------------------------------------------------------|-------------|--|--|--|
| Transmitted command                                            | <b>CUT;</b> |  |  |  |
| Response command                                               | <b>ACK;</b> |  |  |  |

| CUT Switch between video using CUT (enabled when SPLIT is OFF) |        |          |                                                      |
|----------------------------------------------------------------|--------|----------|------------------------------------------------------|
| Transmitted command                                            | CUT:a; |          |                                                      |
|                                                                | a      | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                                                                |        | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                                                                |        | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                                                                |        | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |
|                                                                |        | V-1-4K   | HDMI1-5, STILL1-2, INPUT1-8                          |
| Response command                                               | ACK;   |          |                                                      |

| QATG Get status of Auto Transition |                   |         |  |  |
|------------------------------------|-------------------|---------|--|--|
| Transmitted command                | <b>QATG;</b>      |         |  |  |
| Response command                   | <b>ATG:a;ACK;</b> |         |  |  |
|                                    | <b>a</b>          | OFF, ON |  |  |

## Examples of settings

|                     |                    |                                             |
|---------------------|--------------------|---------------------------------------------|
| Transmitted command | <b>QATG;</b>       | Check whether Auto Transition is being used |
| Response command    | <b>ATG:ON;ACK;</b> | Auto Transition in use                      |

| FRZ Switch between Freeze ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |             |  |  |  |
|-------------------------------------------------------------------|-------------|--|--|--|
| Transmitted command                                               | <b>FRZ;</b> |  |  |  |
| Response command                                                  | <b>ACK;</b> |  |  |  |

## Basic control commands

| FRZ Set Freeze ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |                                   |
|--------------------------------------------------------|-----------------------------------|
| Transmitted command                                    | <b>FRZ:a;</b><br><b>a</b> OFF, ON |
| Response command                                       | <b>ACK;</b>                       |

| QFRZ Get status of Freeze (V-160HD/V-80HD/VR-120HD/VR-6HD) |                                       |
|------------------------------------------------------------|---------------------------------------|
| Transmitted command                                        | <b>QFRZ;</b>                          |
| Response command                                           | <b>FRZ:a;ACK;</b><br><b>a</b> OFF, ON |

| FTB Switch between Output Fade ON/OFF |             |
|---------------------------------------|-------------|
| Transmitted command                   | <b>FTB;</b> |
| Response command                      | <b>ACK;</b> |

| FTB Set Output Fade ON/OFF |                                   |
|----------------------------|-----------------------------------|
| Transmitted command        | <b>FTB:a;</b><br><b>a</b> OFF, ON |
| Response command           | <b>ACK;</b>                       |

| QFTB Get status of Output Fade |                                                        |
|--------------------------------|--------------------------------------------------------|
| Transmitted command            | <b>QFTB;</b>                                           |
| Response command               | <b>FTB:a;ACK;</b><br><b>a</b> OFF, ON, FADEIN, FADEOUT |

| VIS Set Video Input Assign |                                                                          |
|----------------------------|--------------------------------------------------------------------------|
| Transmitted command        | <b>VIS:a,b;</b>                                                          |
|                            | <b>a</b>                                                                 |
|                            | V-160HD INPUT1-20                                                        |
|                            | V-80HD INPUT1-16                                                         |
|                            | VR-120HD INPUT1-8                                                        |
|                            | VR-6HD INPUT1-6                                                          |
|                            | V-1-4K INPUT1-8                                                          |
|                            | <b>b</b>                                                                 |
|                            | V-160HD HDMI1-8, SDI1-8, STILL1-16, N/A                                  |
|                            | V-80HD HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, STATUS1-2, TIME, N/A   |
|                            | VR-120HD HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, STATUS1-2, TIME, N/A |
|                            | VR-6HD HDMI1-6, STILL1-16, V.PLAYER, SRT, N/A                            |
|                            | V-1-4K HDMI, STILL1-2, BLACK, N/A                                        |
| Response command           | <b>ACK;</b>                                                              |

### Examples of settings

|                     |                                                                |
|---------------------|----------------------------------------------------------------|
| Transmitted command | <b>VIS:INPUT1,HDMI1;</b> Set Input Assign of INPUT1 to "HDMI1" |
| Response command    | <b>ACK;</b>                                                    |



| QVIS Get Video Input Assign |                     |          |                                                                 |
|-----------------------------|---------------------|----------|-----------------------------------------------------------------|
| Transmitted command         | <b>QVIS:a;</b>      |          |                                                                 |
|                             | <b>a</b>            | V-160HD  | INPUT1-20                                                       |
|                             |                     | V-80HD   | INPUT1-16                                                       |
|                             |                     | VR-120HD | INPUT1-8                                                        |
|                             |                     | VR-6HD   | INPUT1-6                                                        |
|                             |                     | V-1-4K   | INPUT1-8                                                        |
| Response command            | <b>VIS:a,b;ACK;</b> |          |                                                                 |
|                             | <b>a</b>            | V-160HD  | INPUT1-20                                                       |
|                             |                     | V-80HD   | INPUT1-16                                                       |
|                             |                     | VR-120HD | INPUT1-8                                                        |
|                             |                     | VR-6HD   | INPUT1-6                                                        |
|                             |                     | V-1-4K   | INPUT1-8                                                        |
|                             | <b>b</b>            | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, N/A                                 |
|                             |                     | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, STATUS1-2, TIME, N/A |
|                             |                     | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, STATUS1-2, TIME, N/A |
|                             |                     | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, N/A                          |
|                             |                     | V-1-4K   | HDMI, STILL1-2, BLACK, N/A                                      |

## Examples of settings

|                     |                              |                             |
|---------------------|------------------------------|-----------------------------|
| Transmitted command | <b>QVIS:INPUT1;</b>          | Get Input Assign for INPUT1 |
| Response command    | <b>VIS:INPUT1,HDMI1;ACK;</b> | INPUT1 is HDMI1             |

| VOS Set Video Output Assign |                 |          |                                                                 |
|-----------------------------|-----------------|----------|-----------------------------------------------------------------|
| Transmitted command         | <b>VOS:a,b;</b> |          |                                                                 |
|                             | <b>a</b>        | V-160HD  | HDMI1-3, SDI1-3, USB                                            |
|                             |                 | V-80HD   | HDMI1-3, SDI1-2, USB, STREAM, LCD                               |
|                             |                 | VR-120HD | HDMI1-3, SDI1-3, USB, STREAM, LCD                               |
|                             |                 | VR-6HD   | HDMI1-3, USB, STREAM, LCD                                       |
|                             |                 | V-1-4K   | HDMI3-4                                                         |
|                             | <b>b</b>        | V-160HD  | PGM, SUB, PVW, AUX1-3, DSK1-2, MULTIVIEW, INPUTVIEW1, STILLVIEW |
|                             |                 | V-80HD   | PGM, SUB, PVW, AUX1-2, DSK1, MULTIVIEW, INPUTVIEW1-2, QUADVIEW  |
|                             |                 | VR-120HD | PGM, SUB, PVW, AUX1-3, DSK1-2, MULTIVIEW, INPUTVIEW1, STILLVIEW |
|                             |                 | VR-6HD   | PGM, SUB, PVW, AUX1, DSK1, MULTIVIEW, STILLVIEW                 |
|                             |                 | V-1-4K   | PGM, PVW, INPUT1-8                                              |
| Response command            | <b>ACK;</b>     |          |                                                                 |

| QVOS Get Video Output Assign |                |          |                                   |
|------------------------------|----------------|----------|-----------------------------------|
| Transmitted command          | <b>QVOS:a;</b> |          |                                   |
|                              | <b>a</b>       | V-160HD  | HDMI1-3, SDI1-3, USB              |
|                              |                | V-80HD   | HDMI1-3, SDI1-2, USB, STREAM, LCD |
|                              |                | VR-120HD | HDMI1-3, SDI1-3, USB, STREAM, LCD |
|                              |                | VR-6HD   | HDMI1-3, USB, STREAM, LCD         |
|                              |                | V-1-4K   | HDMI3-4                           |

## Basic control commands

|                  |                     |          |                                                                 |
|------------------|---------------------|----------|-----------------------------------------------------------------|
| Response command | <b>VOS:a,b;ACK;</b> |          |                                                                 |
|                  | <b>a</b>            | V-160HD  | HDMI1-3, SDI1-3, USB                                            |
|                  |                     | V-80HD   | HDMI1-3, SDI1-2, USB, STREAM, LCD                               |
|                  |                     | VR-120HD | HDMI1-3, SDI1-3, USB, STREAM, LCD                               |
|                  |                     | VR-6HD   | HDMI1-3, USB, STREAM, LCD                                       |
|                  |                     | V-1-4K   | HDMI3-4                                                         |
|                  | <b>b</b>            | V-160HD  | PGM, SUB, PVW, AUX1-3, DSK1-2, MULTIVIEW, INPUTVIEW1, STILLVIEW |
|                  |                     | V-80HD   | PGM, SUB, PVW, AUX1-2, DSK1, MULTIVIEW, INPUTVIEW1-2, QUADVIEW  |
|                  |                     | VR-120HD | PGM, SUB, PVW, AUX1-3, DSK1-2, MULTIVIEW, INPUTVIEW1, STILLVIEW |
|                  |                     | VR-6HD   | PGM, SUB, PVW, AUX1, DSK1, MULTIVIEW, STILLVIEW                 |
|                  |                     | V-1-4K   | PGM, PVW, INPUT1-8                                              |

| TRS                 | Set Transition Type |  |           |
|---------------------|---------------------|--|-----------|
| Transmitted command | TRS:a;              |  |           |
|                     | a                   |  | MIX, WIPE |
| Response command    | ACK;                |  |           |

| QTRS                |            | Get Transition Type |  |
|---------------------|------------|---------------------|--|
| Transmitted command | QTRS;      |                     |  |
| Response command    | TRS:a;ACK; |                     |  |
|                     | a          | MIX, WIPE           |  |

| TIM                 | Set Transition Time |          |                                        |
|---------------------|---------------------|----------|----------------------------------------|
| Transmitted command | TIM:a,b;            |          |                                        |
|                     | a                   | V-160HD  | MIX, WIPE, PinP1-4, DSK1-2, OUTPUTFADE |
|                     |                     | V-80HD   | MIX, WIPE, PinP1-2, DSK1, OUTPUTFADE   |
|                     |                     | VR-120HD | MIX, WIPE, PinP1-4, DSK1-2, OUTPUTFADE |
|                     |                     | VR-6HD   | MIX, WIPE, PinP1-2, DSK1, OUTPUTFADE   |
|                     |                     | V-1-4K   | MIX, WIPE, DSK1, OUTPUTFADE            |
|                     | b                   | 0-40     | Transition Time: 0.0-4.0sec            |
| Response command    | ACK;                |          |                                        |

### Examples of settings

|                     |                    |  |  |                                       |
|---------------------|--------------------|--|--|---------------------------------------|
| Transmitted command | <b>TIM:MIX,10;</b> |  |  | Change MIX Transition Time to 1.0 sec |
| Response command    | <b>ACK;</b>        |  |  |                                       |

| QTIM                |         | Get Transition Time |                                        |
|---------------------|---------|---------------------|----------------------------------------|
| Transmitted command | QTIM:a; |                     |                                        |
|                     | a       | V-160HD             | MIX, WIPE, PinP1–4, DSK1–2, OUTPUTFADE |
|                     |         | V-80HD              | MIX, WIPE, PinP1–2, DSK1, OUTPUTFADE   |
|                     |         | VR-120HD            | MIX, WIPE, PinP1–4, DSK1–2, OUTPUTFADE |
|                     |         | VR-6HD              | MIX, WIPE, PinP1–2, DSK1, OUTPUTFADE   |
|                     |         | V-1-4K              | MIX, WIPE, DSK, OUTPUTFADE             |

|                  |                     |          |                                        |
|------------------|---------------------|----------|----------------------------------------|
| Response command | <b>TIM:a,b;ACK;</b> |          |                                        |
|                  | <b>a</b>            | V-160HD  | MIX, WIPE, PinP1-4, DSK1-2, OUTPUTFADE |
|                  |                     | V-80HD   | MIX, WIPE, PinP1-2, DSK1, OUTPUTFADE   |
|                  |                     | VR-120HD | MIX, WIPE, PinP1-4, DSK1-2, OUTPUTFADE |
|                  |                     | VR-6HD   | MIX, WIPE, PinP1-2, DSK1, OUTPUTFADE   |
|                  |                     | V-1-4K   | MIX, WIPE, DSK, OUTPUTFADE             |
|                  | <b>b</b>            | 0-40     | Transition Time: 0.0-4.0sec            |

## Examples of settings

|                     |                        |                                |
|---------------------|------------------------|--------------------------------|
| Transmitted command | <b>QTIM:MIX;</b>       | Query for MIX Transition Time  |
| Response command    | <b>TIM:MIX,10;ACK;</b> | MIX Transition Time is 1.0 sec |

| PIS Set PinP Source (V-160HD/V-80HD/VR-120HD/VR-6HD) |                 |          |                                                      |
|------------------------------------------------------|-----------------|----------|------------------------------------------------------|
| Transmitted command                                  | <b>PIS:a,b;</b> |          |                                                      |
|                                                      | <b>a</b>        | V-160HD  | PinP1-4                                              |
|                                                      |                 | V-80HD   | PinP1-2                                              |
|                                                      |                 | VR-120HD | PinP1-4                                              |
|                                                      |                 | VR-6HD   | PinP1-2                                              |
|                                                      | <b>b</b>        | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                                                      |                 | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                                                      |                 | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                                                      |                 | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |
| Response command                                     | <b>ACK;</b>     |          |                                                      |

| QPIS Get PinP Source (V-160HD/V-80HD/VR-120HD/VR-6HD) |                     |          |                                                      |
|-------------------------------------------------------|---------------------|----------|------------------------------------------------------|
| Transmitted command                                   | <b>QPIS:a;</b>      |          |                                                      |
|                                                       | <b>a</b>            | V-160HD  | PinP1-4                                              |
|                                                       |                     | V-80HD   | PinP1-2                                              |
|                                                       |                     | VR-120HD | PinP1-4                                              |
|                                                       |                     | VR-6HD   | PinP1-2                                              |
| Response command                                      | <b>PIS:a,b;ACK;</b> |          |                                                      |
|                                                       | <b>a</b>            | V-160HD  | PinP1-4                                              |
|                                                       |                     | V-80HD   | PinP1-2                                              |
|                                                       |                     | VR-120HD | PinP1-4                                              |
|                                                       |                     | VR-6HD   | PinP1-2                                              |
|                                                       | <b>b</b>            | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                                                       |                     | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                                                       |                     | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                                                       |                     | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |

| PPS Switch between PinP PGM ON/OFF |               |          |         |
|------------------------------------|---------------|----------|---------|
| Transmitted command                | <b>PPS:a;</b> |          |         |
|                                    | <b>a</b>      | V-160HD  | PinP1-4 |
|                                    |               | V-80HD   | PinP1-2 |
|                                    |               | VR-120HD | PinP1-4 |
|                                    |               | VR-6HD   | PinP1-2 |
|                                    |               | V-1-4K   | PinP1   |
| Response command                   | <b>ACK;</b>   |          |         |

## Basic control commands

| PPS Set PinP PGM ON/OFF |          |          |         |
|-------------------------|----------|----------|---------|
| Transmitted command     | PPS:a,b; |          |         |
|                         | a        | V-160HD  | PinP1-4 |
|                         |          | V-80HD   | PinP1-2 |
|                         |          | VR-120HD | PinP1-4 |
|                         |          | VR-6HD   | PinP1-2 |
|                         |          | V-1-4K   | PinP1   |
|                         | b        | OFF, ON  |         |
| Response command        | ACK;     |          |         |

| QPPS Get status of PinP PGM |                     |          |         |
|-----------------------------|---------------------|----------|---------|
| Transmitted command         | <b>QPPS:a;</b>      |          |         |
|                             | <b>a</b>            | V-160HD  | PinP1-4 |
|                             |                     | V-80HD   | PinP1-2 |
|                             |                     | VR-120HD | PinP1-4 |
|                             |                     | VR-6HD   | PinP1-2 |
|                             |                     | V-1-4K   | PinP1   |
| Response command            | <b>PPS:a,b;ACK;</b> |          |         |
|                             | <b>a</b>            | V-160HD  | PinP1-4 |
|                             |                     | V-80HD   | PinP1-2 |
|                             |                     | VR-120HD | PinP1-4 |
|                             |                     | VR-6HD   | PinP1-2 |
|                             |                     | V-1-4K   | PinP1   |
|                             | <b>b</b>            | OFF, ON  |         |

| PPW Switch between PinP PVW ON/OFF |               |          |         |
|------------------------------------|---------------|----------|---------|
| Transmitted command                | <b>PPW:a;</b> |          |         |
|                                    | <b>a</b>      | V-160HD  | PinP1-4 |
|                                    |               | V-80HD   | PinP1-2 |
|                                    |               | VR-120HD | PinP1-4 |
|                                    |               | VR-6HD   | PinP1-2 |
|                                    |               | V-1-4K   | PinP1   |
| Response command                   | <b>ACK;</b>   |          |         |

| PPW Set PinP PVW ON/OFF |          |          |         |
|-------------------------|----------|----------|---------|
| Transmitted command     | PPW:a,b; |          |         |
|                         | a        | V-160HD  | PinP1-4 |
|                         |          | V-80HD   | PinP1-2 |
|                         |          | VR-120HD | PinP1-4 |
|                         |          | VR-6HD   | PinP1-2 |
|                         |          | V-1-4K   | PinP1   |
|                         | b        | OFF, ON  |         |
| Response command        | ACK;     |          |         |

| QPPW Get PinP PVW status |                     |          |         |
|--------------------------|---------------------|----------|---------|
| Transmitted command      | <b>QPPW:a;</b>      |          |         |
|                          | <b>a</b>            | V-160HD  | PinP1-4 |
|                          |                     | V-80HD   | PinP1-2 |
|                          |                     | VR-120HD | PinP1-4 |
|                          |                     | VR-6HD   | PinP1-2 |
|                          |                     | V-1-4K   | PinP1   |
| Response command         | <b>PPW:a,b;ACK;</b> |          |         |
|                          | <b>a</b>            | V-160HD  | PinP1-4 |
|                          |                     | V-80HD   | PinP1-2 |
|                          |                     | VR-120HD | PinP1-4 |
|                          |                     | VR-6HD   | PinP1-2 |
|                          |                     | V-1-4K   | PinP1   |
|                          | <b>b</b>            | OFF, ON  |         |

| PIP Change PinP Window Position H/V |                   |            |                             |
|-------------------------------------|-------------------|------------|-----------------------------|
| Transmitted command                 | <b>PIP:a,h,v;</b> |            |                             |
|                                     | <b>a</b>          | V-160HD    | PinP1-4                     |
|                                     |                   | V-80HD     | PinP1-2                     |
|                                     |                   | VR-120HD   | PinP1-4                     |
|                                     |                   | VR-6HD     | PinP1-2                     |
|                                     |                   | V-1-4K     | PinP1                       |
|                                     | <b>h</b>          | -500-0-500 | Position H: -50.0-0.0-50.0% |
| Response command                    | <b>v</b>          | -500-0-500 | Position V: -50.0-0.0-50.0% |
|                                     | <b>ACK;</b>       |            |                             |

| QPIP Get PinP Window Position H/V |                       |            |                             |
|-----------------------------------|-----------------------|------------|-----------------------------|
| Transmitted command               | <b>QPIP:a;</b>        |            |                             |
|                                   | <b>a</b>              | V-160HD    | PinP1-4                     |
|                                   |                       | V-80HD     | PinP1-2                     |
|                                   |                       | VR-120HD   | PinP1-4                     |
|                                   |                       | VR-6HD     | PinP1-2                     |
|                                   |                       | V-1-4K     | PinP1                       |
| Response command                  | <b>PIP:a,h,v;ACK;</b> |            |                             |
|                                   | <b>a</b>              | V-160HD    | PinP1-4                     |
|                                   |                       | V-80HD     | PinP1-2                     |
|                                   |                       | VR-120HD   | PinP1-4                     |
|                                   |                       | VR-6HD     | PinP1-2                     |
|                                   | <b>h</b>              | -500-0-500 | Position H: -50.0-0.0-50.0% |
| Response command                  | <b>v</b>              | -500-0-500 | Position V: -50.0-0.0-50.0% |

| DSK Switch between DSK PGM ON/OFF |               |          |        |
|-----------------------------------|---------------|----------|--------|
| Transmitted command               | <b>DSK:a;</b> |          |        |
|                                   | <b>a</b>      | V-160HD  | DSK1-2 |
|                                   |               | V-80HD   | DSK1   |
|                                   |               | VR-120HD | DSK1-2 |
|                                   |               | VR-6HD   | DSK1   |
|                                   |               | V-1-4K   | DSK1   |
| Response command                  | <b>ACK;</b>   |          |        |

## Basic control commands

| DSK                 |          | Set DSK PGM ON/OFF |        |
|---------------------|----------|--------------------|--------|
| Transmitted command | DSK:a,b; |                    |        |
|                     | a        | V-160HD            | DSK1-2 |
|                     |          | V-80HD             | DSK1   |
|                     |          | VR-120HD           | DSK1-2 |
|                     |          | VR-6HD             | DSK1   |
|                     |          | V-1-4K             | DSK1   |
|                     | b        | OFF, ON            |        |
| Response command    | ACK;     |                    |        |

| QDSK Get status of DSK PGM |                     |          |        |
|----------------------------|---------------------|----------|--------|
| Transmitted command        | <b>QDSK:a;</b>      |          |        |
|                            | <b>a</b>            | V-160HD  | DSK1-2 |
|                            |                     | V-80HD   | DSK1   |
|                            |                     | VR-120HD | DSK1-2 |
|                            |                     | VR-6HD   | DSK1   |
|                            |                     | V-1-4K   | DSK1   |
| Response command           | <b>DSK:a,b;ACK;</b> |          |        |
|                            | <b>a</b>            | V-160HD  | DSK1-2 |
|                            |                     | V-80HD   | DSK1   |
|                            |                     | VR-120HD | DSK1-2 |
|                            |                     | VR-6HD   | DSK1   |
|                            |                     | V-1-4K   | DSK1   |
|                            | <b>b</b>            | OFF, ON  |        |

| DVW Switch between DSK PVW ON/OFF |               |          |        |
|-----------------------------------|---------------|----------|--------|
| Transmitted command               | <b>DVW:a;</b> |          |        |
|                                   | <b>a</b>      | V-160HD  | DSK1-2 |
|                                   |               | V-80HD   | DSK1   |
|                                   |               | VR-120HD | DSK1-2 |
|                                   |               | VR-6HD   | DSK1   |
|                                   |               | V-1-4K   | DSK1   |
| Response command                  | <b>ACK;</b>   |          |        |

| DVW Set DSK PVW ON/OFF |          |          |        |
|------------------------|----------|----------|--------|
| Transmitted command    | DVW:a,b; |          |        |
|                        | a        | V-160HD  | DSK1-2 |
|                        |          | V-80HD   | DSK1   |
|                        |          | VR-120HD | DSK1-2 |
|                        |          | VR-6HD   | DSK1   |
|                        |          | V-1-4K   | DSK1   |
|                        | b        | OFF, ON  |        |
| Response command       | ACK;     |          |        |

| QDVW                |              | Get status of DSK PVW |        |
|---------------------|--------------|-----------------------|--------|
| Transmitted command | QDVW:a;      |                       |        |
|                     | a            | V-160HD               | DSK1-2 |
|                     |              | V-80HD                | DSK1   |
|                     |              | VR-120HD              | DSK1-2 |
|                     |              | VR-6HD                | DSK1   |
|                     |              | V-1-4K                | DSK1   |
| Response command    | DVW:a,b;ACK; |                       |        |
|                     | a            | V-160HD               | DSK1-2 |
|                     |              | V-80HD                | DSK1   |
|                     |              | VR-120HD              | DSK1-2 |
|                     |              | VR-6HD                | DSK1   |
|                     |              | V-1-4K                | DSK1   |
|                     | b            | OFF, ON               |        |

| DSS                 | Change DSK Fill Source |          |                                                      |
|---------------------|------------------------|----------|------------------------------------------------------|
| Transmitted command | DSS:a,b;               |          |                                                      |
|                     | a                      | V-160HD  | DSK1-2                                               |
|                     |                        | V-80HD   | DSK1                                                 |
|                     |                        | VR-120HD | DSK1-2                                               |
|                     |                        | VR-6HD   | DSK1                                                 |
|                     |                        | V-1-4K   | DSK1                                                 |
|                     | b                      | V-160HD  | HDMI1-8, SDI1-8, STILL1-16, INPUT1-20                |
|                     |                        | V-80HD   | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT, INPUT1-16 |
|                     |                        | VR-120HD | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-8  |
|                     |                        | VR-6HD   | HDMI1-6, STILL1-16, V.PLAYER, SRT, INPUT1-6          |
|                     |                        | V-1-4K   | HDMI1-5, STILL1-2, INPUT1-8                          |
| Response command    | ACK;                   |          |                                                      |

| QDSS                |                | Get DSK Fill Source |                                           |
|---------------------|----------------|---------------------|-------------------------------------------|
| Transmitted command | QDSS:a;        |                     |                                           |
|                     | a              | V-160HD             | DSK1-2                                    |
|                     |                | V-80HD              | DSK1                                      |
|                     |                | VR-120HD            | DSK1-2                                    |
|                     |                | VR-6HD              | DSK1                                      |
|                     |                | V-1-4K              | DSK1                                      |
| Response command    | DSS:a,b,c;ACK; |                     |                                           |
|                     | a              | V-160HD             | DSK1-2                                    |
|                     |                | V-80HD              | DSK1                                      |
|                     |                | VR-120HD            | DSK1-2                                    |
|                     |                | VR-6HD              | DSK1                                      |
|                     |                | V-1-4K              | DSK1                                      |
|                     | b              | V-160HD             | HDMI1-8, SDI1-8, STILL1-16                |
|                     |                | V-80HD              | HDMI1-4, SDI1-4, STILL1-32, V.PLAYER, SRT |
|                     |                | VR-120HD            | HDMI1-6, SDI1-6, STILL1-16, V.PLAYER, SRT |
|                     |                | VR-6HD              | HDMI1-6, STILL1-16, V.PLAYER, SRT         |
|                     |                | V-1-4K              | HDMI1-5, STILL1-2, INPUT1-8               |
|                     | c              | V-160HD             | INPUT1-20                                 |
|                     |                | V-80HD              | INPUT1-16                                 |
|                     |                | VR-120HD            | INPUT1-8                                  |
|                     |                | VR-6HD              | INPUT1-6                                  |
|                     |                | V-1-4K              | INPUT1-8                                  |

## Basic control commands

| KYL                 | Change DSK Level |          |        |
|---------------------|------------------|----------|--------|
| Transmitted command | KYL:a,b;         |          |        |
|                     | a                | V-160HD  | DSK1-2 |
|                     |                  | V-80HD   | DSK1   |
|                     |                  | VR-120HD | DSK1-2 |
|                     |                  | VR-6HD   | DSK1   |
|                     |                  | V-1-4K   | DSK1   |
|                     | b                | 0-255    |        |
| Response command    | ACK;             |          |        |

| QKYL Get DSK Level  |                     |          |        |
|---------------------|---------------------|----------|--------|
| Transmitted command | <b>QKYL:a;</b>      |          |        |
|                     | <b>a</b>            | V-160HD  | DSK1-2 |
|                     |                     | V-80HD   | DSK1   |
|                     |                     | VR-120HD | DSK1-2 |
|                     |                     | VR-6HD   | DSK1   |
|                     |                     | V-1-4K   | DSK1   |
| Response command    | <b>KYL:a,b;ACK;</b> |          |        |
|                     | <b>a</b>            | V-160HD  | DSK1-2 |
|                     |                     | V-80HD   | DSK1   |
|                     |                     | VR-120HD | DSK1-2 |
|                     |                     | VR-6HD   | DSK1   |
|                     |                     | V-1-4K   | DSK1   |
|                     | <b>b</b>            | 0-255    |        |

| KYG                 | Change DSK Gain |          |        |
|---------------------|-----------------|----------|--------|
| Transmitted command | KYG:a,b;        |          |        |
|                     | a               | V-160HD  | DSK1-2 |
|                     |                 | V-80HD   | DSK1   |
|                     |                 | VR-120HD | DSK1-2 |
|                     |                 | VR-6HD   | DSK1   |
|                     |                 | V-1-4K   | DSK1   |
|                     | b               | 0-255    |        |
| Response command    | ACK;            |          |        |

| QKYG Get DSK Gain   |                     |          |        |
|---------------------|---------------------|----------|--------|
| Transmitted command | <b>QKYG:a;</b>      |          |        |
|                     | <b>a</b>            | V-160HD  | DSK1-2 |
|                     |                     | V-80HD   | DSK1   |
|                     |                     | VR-120HD | DSK1-2 |
|                     |                     | VR-6HD   | DSK1   |
|                     |                     | V-1-4K   | DSK1   |
| Response command    | <b>KYG:a,b;ACK;</b> |          |        |
|                     | <b>a</b>            | V-160HD  | DSK1-2 |
|                     |                     | V-80HD   | DSK1   |
|                     |                     | VR-120HD | DSK1-2 |
|                     |                     | VR-6HD   | DSK1   |
|                     |                     | V-1-4K   | DSK1   |
|                     | <b>b</b>            | 0-255    |        |



| SPS Switch between SPLIT ON/OFF |               |          |          |
|---------------------------------|---------------|----------|----------|
| Transmitted command             | <b>SPS:a;</b> |          |          |
|                                 | <b>a</b>      | V-160HD  | SPLIT1-2 |
|                                 |               | V-80HD   | SPLIT1-2 |
|                                 |               | VR-120HD | SPLIT1-2 |
|                                 |               | VR-6HD   | SPLIT1   |
|                                 |               | V-1-4K   | SPLIT1   |
| Response command                | <b>ACK;</b>   |          |          |

| SPS                 |          |          |         | Set SPLIT ON/OFF |  |  |  |
|---------------------|----------|----------|---------|------------------|--|--|--|
| Transmitted command | SPS:a,b; |          |         |                  |  |  |  |
|                     | a        | V-160HD  |         | SPLIT1-2         |  |  |  |
|                     |          | V-80HD   |         | SPLIT1-2         |  |  |  |
|                     |          | VR-120HD |         | SPLIT1-2         |  |  |  |
|                     |          | VR-6HD   |         | SPLIT1           |  |  |  |
|                     |          | V-1-4K   |         | SPLIT1           |  |  |  |
|                     | b        |          | OFF, ON |                  |  |  |  |
| Response command    | ACK;     |          |         |                  |  |  |  |

| QSPS Get status of SPLIT |                     |          |          |
|--------------------------|---------------------|----------|----------|
| Transmitted command      | <b>QSPS:a;</b>      |          |          |
|                          | <b>a</b>            | V-160HD  | SPLIT1-2 |
|                          |                     | V-80HD   | SPLIT1-2 |
|                          |                     | VR-120HD | SPLIT1-2 |
|                          |                     | VR-6HD   | SPLIT1   |
|                          |                     | V-1-4K   | SPLIT1   |
| Response command         | <b>SPS:a,b;ACK;</b> |          |          |
|                          | <b>a</b>            | V-160HD  | SPLIT1-2 |
|                          |                     | V-80HD   | SPLIT1-2 |
|                          |                     | VR-120HD | SPLIT1-2 |
|                          |                     | VR-6HD   | SPLIT1   |
|                          |                     | V-1-4K   | SPLIT1   |
|                          | <b>b</b>            | OFF, ON  |          |

| SPT Set SPLIT PGM/A-Center, PST/B-Center |                   |            |                               |
|------------------------------------------|-------------------|------------|-------------------------------|
| Transmitted command                      | <b>SPT:a,b,c;</b> |            |                               |
|                                          | <b>a</b>          | V-160HD    | SPLIT1-2                      |
|                                          |                   | V-80HD     | SPLIT1-2                      |
|                                          |                   | VR-120HD   | SPLIT1-2                      |
|                                          |                   | VR-6HD     | SPLIT1                        |
|                                          |                   | V-1-4K     | SPLIT1                        |
|                                          | <b>b</b>          | -500-0-500 | PGM/A-Center: -50.0-0.0-50.0% |
|                                          | <b>c</b>          | -500-0-500 | PST/B-Center: -50.0-0.0-50.0% |
| Response command                         | <b>ACK;</b>       |            |                               |

## Basic control commands

| SPT Set SPLIT PGM/A-Center, PST/B-Center, Center Position |                     |          |            |                                  |
|-----------------------------------------------------------|---------------------|----------|------------|----------------------------------|
| Transmitted command                                       | <b>SPT:a,b,c,d;</b> |          |            |                                  |
|                                                           | <b>a</b>            | V-160HD  | SPLIT1-2   |                                  |
|                                                           |                     | V-80HD   | SPLIT1-2   |                                  |
|                                                           |                     | VR-120HD | SPLIT1-2   |                                  |
|                                                           |                     | VR-6HD   | SPLIT1     |                                  |
|                                                           |                     | V-1-4K   | SPLIT1     |                                  |
|                                                           | <b>b</b>            |          | -500-0-500 | PGM/A-Center: -50.0-0.0-50.0%    |
| Response command                                          | <b>c</b>            |          | -500-0-500 | PST/B-Center: -50.0-0.0-50.0%    |
|                                                           | <b>d</b>            |          | -500-0-500 | Center Position: -50.0-0.0-50.0% |
|                                                           | <b>ACK;</b>         |          |            |                                  |

| QSPT Get SPLIT PGM/A-Center, PST/B-Center, Center Position |                         |          |            |                                  |
|------------------------------------------------------------|-------------------------|----------|------------|----------------------------------|
| Transmitted command                                        | <b>QSPT:a;</b>          |          |            |                                  |
|                                                            | <b>a</b>                | V-160HD  | SPLIT1-2   |                                  |
|                                                            |                         | V-80HD   | SPLIT1-2   |                                  |
|                                                            |                         | VR-120HD | SPLIT1-2   |                                  |
|                                                            |                         | VR-6HD   | SPLIT1     |                                  |
|                                                            |                         | V-1-4K   | SPLIT1     |                                  |
| Response command                                           | <b>SPT:a,b,c,d;ACK;</b> |          |            |                                  |
|                                                            | <b>a</b>                | V-160HD  | SPLIT1-2   |                                  |
|                                                            |                         | V-80HD   | SPLIT1-2   |                                  |
|                                                            |                         | VR-120HD | SPLIT1-2   |                                  |
|                                                            |                         | VR-6HD   | SPLIT1     |                                  |
|                                                            |                         | V-1-4K   | SPLIT1     |                                  |
|                                                            | <b>b</b>                |          | -500-0-500 | PGM/A-Center: -50.0-0.0-50.0%    |
|                                                            | <b>c</b>                |          | -500-0-500 | PST/B-Center: -50.0-0.0-50.0%    |
|                                                            | <b>d</b>                |          | -500-0-500 | Center Position: -50.0-0.0-50.0% |

| STO Set STILL IMAGE OUTPUT (V-160HD/V-80HD/VR-120HD/VR-6HD) |               |          |                               |  |
|-------------------------------------------------------------|---------------|----------|-------------------------------|--|
| Transmitted command                                         | <b>STO:a;</b> |          |                               |  |
|                                                             | <b>a</b>      | V-160HD  | OFF, STILL1-16                |  |
|                                                             |               | V-80HD   | OFF, STILL1-32, V.PLAYER, SRT |  |
|                                                             |               | VR-120HD | OFF, STILL1-16, V.PLAYER, SRT |  |
|                                                             |               | VR-6HD   | OFF, STILL1-16, V.PLAYER, SRT |  |
| Response command                                            | <b>ACK;</b>   |          |                               |  |

| QSTO Get STILL IMAGE OUTPUT (V-160HD/V-80HD/VR-120HD/VR-6HD) |                   |          |                               |  |
|--------------------------------------------------------------|-------------------|----------|-------------------------------|--|
| Transmitted command                                          | <b>QSTO;</b>      |          |                               |  |
| Response command                                             | <b>STO:a;ACK;</b> |          |                               |  |
|                                                              | <b>a</b>          | V-160HD  | OFF, STILL1-16                |  |
|                                                              |                   | V-80HD   | OFF, STILL1-32, V.PLAYER, SRT |  |
|                                                              |                   | VR-120HD | OFF, STILL1-16, V.PLAYER, SRT |  |
|                                                              |                   | VR-6HD   | OFF, STILL1-16, V.PLAYER, SRT |  |

| ROI SW      Switch the ROI mode ON/OFF (V-1-4K) |                  |
|-------------------------------------------------|------------------|
| Transmitted command                             | <b>ROI SW:a;</b> |
|                                                 | <b>a</b> OFF, ON |
| Response command                                | <b>ACK;</b>      |

| QROI SW      Get status of ROI mode (V-1-4K) |                      |
|----------------------------------------------|----------------------|
| Transmitted command                          | <b>QROI SW;</b>      |
| Response command                             | <b>ROI SW:a;ACK;</b> |
|                                              | <b>a</b> OFF, ON     |

| ROI ZM      Change ROI ZOOM (V-1-4K) |                                            |
|--------------------------------------|--------------------------------------------|
| Transmitted command                  | <b>ROI ZM:a,b;</b>                         |
|                                      | <b>a</b> ROI1-4                            |
|                                      | <b>b</b> 1000-4000      Zoom: 100.0-400.0% |
| Response command                     | <b>ACK;</b>                                |

| QROI ZM      Get ROI ZOOM (V-1-4K) |                                            |
|------------------------------------|--------------------------------------------|
| Transmitted command                | <b>QROI ZM:a;</b>                          |
|                                    | <b>a</b> ROI1-4                            |
| Response command                   | <b>ROI ZM:a,b;ACK;</b>                     |
|                                    | <b>a</b> ROI1-4                            |
|                                    | <b>b</b> 1000-4000      Zoom: 100.0-400.0% |

| ROI PT      Change ROI POSITION H/V (V-1-4K) |                                                   |
|----------------------------------------------|---------------------------------------------------|
| Transmitted command                          | <b>ROI PT:a,b,c;</b>                              |
|                                              | <b>a</b> ROI1-4                                   |
|                                              | <b>b</b> -1920-1920      Position H: -1920-0-1920 |
|                                              | <b>c</b> -1080-1080      Position V: -1080-0-1080 |
| Response command                             | <b>ACK;</b>                                       |

| QROI PT      Get ROI POSITION H/V (V-1-4K) |                                                   |
|--------------------------------------------|---------------------------------------------------|
| Transmitted command                        | <b>QROI PT:a;</b>                                 |
|                                            | <b>a</b> ROI1-4                                   |
| Response command                           | <b>ROI PT:a,b,c;ACK;</b>                          |
|                                            | <b>a</b> ROI1-4                                   |
|                                            | <b>b</b> -1920-1920      Position H: -1920-0-1920 |
|                                            | <b>c</b> -1080-1080      Position V: -1080-0-1080 |

## AUDIO

| AOS                 |          | Set Audio Output Assign |                                                         |                                                                        |  |
|---------------------|----------|-------------------------|---------------------------------------------------------|------------------------------------------------------------------------|--|
| Transmitted command | AOS:a,b; |                         |                                                         |                                                                        |  |
|                     | a        | V-160HD                 | HDMI1-3, SDI1-3, XLR1, RCA1, USB, PHONES                |                                                                        |  |
|                     |          | V-80HD                  | HDMI1-3, SDI1-2, XLR1, USB, STREAM, REC, PHONES         |                                                                        |  |
|                     |          | VR-120HD                | HDMI1-3, SDI1-3, XLR1-2, RCA1, USB, STREAM, REC, PHONES |                                                                        |  |
|                     |          | VR-6HD                  | HDMI1-3, XLR1, RCA1, USB, STREAM, REC, PHONES           |                                                                        |  |
|                     |          | V-1-4K                  | HDMI1-5, XLR1, USB, PHONES                              |                                                                        |  |
|                     | b        | V-160HD                 | AUTO, MAIN, AUX1-3                                      | * This can be set when parameter a is HDMI1-3, SDI1-3 or USB.          |  |
|                     |          |                         | MAIN, AUX1-3                                            | * This can be set when parameter a is XLR1, RCA1 or PHONES.            |  |
|                     |          | V-80HD                  | AUTO, MAIN, AUX1-2, MONITOR                             | * This can be set when parameter a is HDMI1-3, SDI1-2, USB or STREAM.  |  |
|                     |          |                         | MAIN, AUX1-2, MONITOR                                   | * This can be set when parameter a is XLR1 or REC.                     |  |
|                     |          |                         | MAIN, AUX1-2                                            | * This can be set when parameter a is PHONES.                          |  |
|                     |          | VR-120HD                | MAIN, AUX1-3, MONITOR                                   | * This can be set when parameter a is HDMI1-3, SDI1-3, XLR1-2 or RCA1. |  |
|                     |          |                         | MAIN, AUX1-3                                            | * This can be set when parameter a is USB, STREAM, REC or PHONES.      |  |
|                     |          | VR-6HD                  | MAIN, AUX1-2, MONITOR                                   | * This can be set when parameter a is HDMI1-3, XLR1 or RCA1.           |  |
|                     |          |                         | MAIN, AUX1-2                                            | * This can be set when parameter a is USB, STREAM, REC or PHONES.      |  |
|                     |          | V-1-4K                  | MAIN, AUX1-2, MONITOR                                   | * This can be set when parameter a is HDMI1-5, XLR1 or USB.            |  |
|                     |          |                         | MAIN, AUX1-2                                            | * This can be set when parameter a is PHONES.                          |  |
|                     |          | Response command        | ACK;                                                    |                                                                        |  |

| QAOS                    |         |          |                                                         |
|-------------------------|---------|----------|---------------------------------------------------------|
| Get Audio Output Assign |         |          |                                                         |
| Transmitted command     | QAOS:a; |          |                                                         |
|                         | a       | V-160HD  | HDMI1-3, SDI1-3, XLR1, RCA1, USB, PHONES                |
|                         |         | V-80HD   | HDMI1-3, SDI1-2, XLR1, USB, STREAM, REC, PHONES         |
|                         |         | VR-120HD | HDMI1-3, SDI1-3, XLR1-2, RCA1, USB, STREAM, REC, PHONES |
|                         |         | VR-6HD   | HDMI1-3, XLR1, RCA1, USB, STREAM, REC, PHONES           |
|                         |         | V-1-4K   | HDMI1-5, XLR1, USB, PHONES                              |

| Response command | <b>AOS:a,b;ACK;</b> |          |                                                         |                                                                     |
|------------------|---------------------|----------|---------------------------------------------------------|---------------------------------------------------------------------|
|                  | <b>a</b>            | V-160HD  | HDMI1-3, SDI1-3, XLR1, RCA1, USB, PHONES                |                                                                     |
|                  |                     | V-80HD   | HDMI1-3, SDI1-2, XLR1, USB, STREAM, REC, PHONES         |                                                                     |
|                  |                     | VR-120HD | HDMI1-3, SDI1-3, XLR1-2, RCA1, USB, STREAM, REC, PHONES |                                                                     |
|                  |                     | VR-6HD   | HDMI1-3, XLR1, RCA1, USB, STREAM, REC, PHONES           |                                                                     |
|                  |                     | V-1-4K   | HDMI1-5, XLR1, USB, PHONES                              |                                                                     |
|                  | <b>b</b>            | V-160HD  | AUTO, MAIN, AUX1-3                                      | * This is sent when parameter a is HDMI1-3, SDI1-3 or USB.          |
|                  |                     |          | MAIN, AUX1-3                                            | * This is sent when parameter a is XLR1, RCA1 or PHONES.            |
|                  |                     | V-80HD   | AUTO, MAIN, AUX1-2, MONITOR                             | * This is sent when parameter a is HDMI1-3, SDI1-2, USB or STREAM.  |
|                  |                     |          | MAIN, AUX1-2, MONITOR                                   | * This is sent when parameter a is XLR1 or REC.                     |
|                  |                     |          | MAIN, AUX1-2                                            | * This is sent when parameter a is PHONES.                          |
|                  |                     | VR-120HD | MAIN, AUX1-3, MONITOR                                   | * This is sent when parameter a is HDMI1-3, SDI1-3, XLR1-2 or RCA1. |
|                  |                     |          | MAIN, AUX1-3                                            | * This is sent when parameter a is USB, STREAM, REC or PHONES.      |
|                  |                     | VR-6HD   | MAIN, AUX1-2, MONITOR                                   | * This is sent when parameter a is HDMI1-3, XLR1 or RCA1.           |
|                  |                     |          | MAIN, AUX1-2                                            | * This is sent when parameter a is USB, STREAM, REC or PHONES.      |
|                  |                     | V-1-4K   | MAIN, AUX1-2, MONITOR                                   | * This can be set when parameter a is HDMI1-5, XLR1 or USB.         |
|                  |                     |          | MAIN, AUX1-2                                            | * This can be set when parameter a is PHONES.                       |

#### IAL Set Audio Input Level

| Transmitted command | IAL:a,b;         |                  |                                                                          |                               |
|---------------------|------------------|------------------|--------------------------------------------------------------------------|-------------------------------|
|                     | a                | V-160HD          | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |                               |
|                     |                  | V-80HD           | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |                               |
|                     |                  | VR-120HD         | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |                               |
|                     |                  | VR-6HD           | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |                               |
|                     |                  | V-1-4K           | XLR1, USB, HDMI1-5                                                       |                               |
|                     | b                | -INF, -800-0-100 |                                                                          | Level: -INF, -80.0-0.0-10.0dB |
|                     | Response command | ACK;             |                                                                          |                               |

#### Examples of settings

| Transmitted command | <b>IAL:XLR1,-60;</b> |  |  | Set Audio Input Level of XLR1 to -6.0 dB |
|---------------------|----------------------|--|--|------------------------------------------|
| Response command    | <b>ACK;</b>          |  |  |                                          |

#### QIAL Get Audio Input Level

| Transmitted command | QIAL:a;      |                  |                                                                          |
|---------------------|--------------|------------------|--------------------------------------------------------------------------|
|                     | a            | V-160HD          | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                     |              | V-80HD           | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                     |              | VR-120HD         | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                     |              | VR-6HD           | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                     |              | V-1-4K           | XLR1, USB, HDMI1-5                                                       |
| Response command    | IAL:a,b;ACK; |                  |                                                                          |
|                     | a            | V-160HD          | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                     |              | V-80HD           | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                     |              | VR-120HD         | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                     |              | VR-6HD           | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                     |              | V-1-4K           | XLR1, USB, HDMI1-5                                                       |
|                     | b            | -INF, -800-0-100 |                                                                          |

### Examples of settings

|                     |                          |                                      |
|---------------------|--------------------------|--------------------------------------|
| Transmitted command | <b>QIAL:XLR1;</b>        | Check the Audio Input Level of XLR1  |
| Response command    | <b>IAL:XLR1,-60;ACK;</b> | Audio Input Level of XLR1 is -6.0 dB |

| IAM Switch between Audio Input Mute ON/OFF |               |          |                                                                          |
|--------------------------------------------|---------------|----------|--------------------------------------------------------------------------|
| Transmitted command                        | <b>IAM:a;</b> |          |                                                                          |
|                                            | <b>a</b>      | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                            |               | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                            |               | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                            |               | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                            |               | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command                           | <b>ACK;</b>   |          |                                                                          |

| IAM Set Audio Input Mute ON/OFF |          |          |                                                                          |
|---------------------------------|----------|----------|--------------------------------------------------------------------------|
| Transmitted command             | IAM:a,b; |          |                                                                          |
|                                 | a        | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                 |          | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                 |          | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                 |          | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                 |          | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                 | b        | OFF, ON  |                                                                          |
| Response command                | ACK;     |          |                                                                          |

| QIAM Get status of Audio Input Mute |                     |          |                                                                          |
|-------------------------------------|---------------------|----------|--------------------------------------------------------------------------|
| Transmitted command                 | <b>QIAM:a;</b>      |          |                                                                          |
|                                     | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                     |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                     |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                     |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                     |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command                    | <b>IAM:a,b;ACK;</b> |          |                                                                          |
|                                     | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                     |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                     |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                     |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                     |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                     | <b>b</b>            | OFF, ON  |                                                                          |

| IAS Switch between Audio Input Solo ON/OFF |               |          |                                                                          |
|--------------------------------------------|---------------|----------|--------------------------------------------------------------------------|
| Transmitted command                        | <b>IAS:a;</b> |          |                                                                          |
|                                            | <b>a</b>      | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                            |               | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                            |               | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                            |               | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                            |               | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command                           | <b>ACK;</b>   |          |                                                                          |

| IAS Set Audio Input Solo ON/OFF |          |          |                                                                          |
|---------------------------------|----------|----------|--------------------------------------------------------------------------|
| Transmitted command             | IAS:a,b; |          |                                                                          |
|                                 | a        | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                 |          | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                 |          | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                 |          | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                 |          | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                 | b        | OFF, ON  |                                                                          |
| Response command                | ACK;     |          |                                                                          |

| QIAS Get status of Audio Input Solo |                     |          |                                                                          |
|-------------------------------------|---------------------|----------|--------------------------------------------------------------------------|
| Transmitted command                 | <b>QIAS:a;</b>      |          |                                                                          |
|                                     | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                     |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                     |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                     |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                     |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command                    | <b>IAS:a,b;ACK;</b> |          |                                                                          |
|                                     | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                     |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                     |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                     |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                     |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                     | <b>b</b>            | OFF, ON  |                                                                          |

| ADT Set Audio Input Delay Time |                 |          |                                                                          |
|--------------------------------|-----------------|----------|--------------------------------------------------------------------------|
| Transmitted command            | <b>ADT:a,b;</b> |          |                                                                          |
|                                | <b>a</b>        | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                |                 | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                |                 | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                |                 | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                |                 | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command               | <b>b</b>        | 0-5000   | 0.0-500.0msec                                                            |
|                                | <b>ACK;</b>     |          |                                                                          |

| QADT Get Audio Input Delay Time |                     |          |                                                                          |
|---------------------------------|---------------------|----------|--------------------------------------------------------------------------|
| Transmitted command             | <b>QADT:a;</b>      |          |                                                                          |
|                                 | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                 |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                 |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                 |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                 |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command                | <b>ADT:a,b;ACK;</b> |          |                                                                          |
|                                 | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                 |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                 |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                 |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                 |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                 | <b>b</b>            | 0-5000   | 0.0-500.0msec                                                            |

## Basic control commands

| HPF Set High Pass Filter ON/OFF |          |          |                                                                          |
|---------------------------------|----------|----------|--------------------------------------------------------------------------|
| Transmitted command             | HPF:a,b; |          |                                                                          |
|                                 | a        | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                 |          | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                 |          | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                 |          | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                 |          | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                 | b        | OFF, ON  |                                                                          |
| Response command                | ACK;     |          |                                                                          |

| QHPF Get status of High Pass Filter |                     |          |                                                                          |
|-------------------------------------|---------------------|----------|--------------------------------------------------------------------------|
| Transmitted command                 | <b>QHPF:a;</b>      |          |                                                                          |
|                                     | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                     |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                     |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                     |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                     |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command                    | <b>HPF:a,b;ACK;</b> |          |                                                                          |
|                                     | <b>a</b>            | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                                     |                     | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                                     |                     | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                                     |                     | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                                     |                     | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                                     | <b>b</b>            | OFF, ON  |                                                                          |

| GATE                |           |          |                                                                          | Set GATE ON/OFF |  |  |  |
|---------------------|-----------|----------|--------------------------------------------------------------------------|-----------------|--|--|--|
| Transmitted command | GATE:a,b; |          |                                                                          |                 |  |  |  |
|                     | a         | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |                 |  |  |  |
|                     |           | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |                 |  |  |  |
|                     |           | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |                 |  |  |  |
|                     |           | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |                 |  |  |  |
|                     |           | V-1-4K   | XLR1, USB, HDMI1-5                                                       |                 |  |  |  |
|                     | b         | OFF, ON  |                                                                          |                 |  |  |  |
| Response command    | ACK;      |          |                                                                          |                 |  |  |  |

| QGATE Get status of GATE |                      |          |                                                                          |
|--------------------------|----------------------|----------|--------------------------------------------------------------------------|
| Transmitted command      | <b>QGATE:a;</b>      |          |                                                                          |
|                          | <b>a</b>             | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                          |                      | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                          |                      | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                          |                      | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                          |                      | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
| Response command         | <b>GATE:a,b;ACK;</b> |          |                                                                          |
|                          | <b>a</b>             | V-160HD  | XLR1-2, RCA1, USB, BLUETOOTH, HDMI1-8, SDI1-8                            |
|                          |                      | V-80HD   | XLR1-2, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-4, SDI1-4, V.PLAYER, SRT   |
|                          |                      | VR-120HD | XLR1-6, RCA1-2, USB, BLUETOOTH, A.PLAYER, HDMI1-6, SDI1-6, V.PLAYER, SRT |
|                          |                      | VR-6HD   | XLR1-6, RCA1, USB, BLUETOOTH, A.PLAYER, HDMI1-6, V.PLAYER, SRT           |
|                          |                      | V-1-4K   | XLR1, USB, HDMI1-5                                                       |
|                          | <b>b</b>             | OFF, ON  |                                                                          |



| STLK Switch between Stereo Link ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |                |          |        |
|-------------------------------------------------------------------------|----------------|----------|--------|
| Transmitted command                                                     | <b>STLK:a;</b> |          |        |
|                                                                         | <b>a</b>       | V-160HD  | XLR1-2 |
|                                                                         |                | V-80HD   | XLR1-2 |
|                                                                         |                | VR-120HD | XLR1-6 |
|                                                                         |                | VR-6HD   | XLR1-6 |
| Response command                                                        | <b>ACK;</b>    |          |        |

| STLK Set Stereo Link ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |           |          |        |
|--------------------------------------------------------------|-----------|----------|--------|
| Transmitted command                                          | STLK:a,b; |          |        |
|                                                              | a         | V-160HD  | XLR1-2 |
|                                                              |           | V-80HD   | XLR1-2 |
|                                                              |           | VR-120HD | XLR1-6 |
|                                                              |           | VR-6HD   | XLR1-6 |
|                                                              | b         | OFF, ON  |        |
| Response command                                             | ACK;      |          |        |

| QSTLK Get status of Stereo Link (V-160HD/V-80HD/VR-120HD/VR-6HD) |                      |          |        |
|------------------------------------------------------------------|----------------------|----------|--------|
| Transmitted command                                              | <b>QSTLK:a;</b>      |          |        |
|                                                                  | <b>a</b>             | V-160HD  | XLR1-2 |
|                                                                  |                      | V-80HD   | XLR1-2 |
|                                                                  |                      | VR-120HD | XLR1-6 |
|                                                                  |                      | VR-6HD   | XLR1-6 |
| Response command                                                 | <b>STLK:a,b;ACK;</b> |          |        |
|                                                                  | <b>a</b>             | V-160HD  | XLR1-2 |
|                                                                  |                      | V-80HD   | XLR1-2 |
|                                                                  |                      | VR-120HD | XLR1-6 |
|                                                                  |                      | VR-6HD   | XLR1-6 |
|                                                                  | <b>b</b>             | OFF, ON  |        |

| VOCH Switch between Voice Changer ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |         |        |  |
|---------------------------------------------------------------------------|---------|--------|--|
| Transmitted command                                                       | VOCH:a; |        |  |
|                                                                           | a       | XLR1-2 |  |
| Response command                                                          | ACK;    |        |  |

| VOCH                |           | Set Voice Changer ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |  |
|---------------------|-----------|-----------------------------------------------------------|--|
| Transmitted command | VOCH:a,b; |                                                           |  |
|                     | a         | XLR1-2                                                    |  |
|                     | b         | OFF, ON                                                   |  |
| Response command    | ACK;      |                                                           |  |

| QVOCH Get status of Voice Changer (V-160HD/V-80HD/VR-120HD/VR-6HD) |                 |        |  |
|--------------------------------------------------------------------|-----------------|--------|--|
| Transmitted command                                                | <b>QVOCH:a;</b> |        |  |
|                                                                    | <b>a</b>        | XLR1-2 |  |

## Basic control commands

|                  |                      |         |                   |
|------------------|----------------------|---------|-------------------|
| Response command | <b>VOCH:a,b;ACK;</b> |         |                   |
|                  | <b>a</b>             | V-160HD | MAIN, AUX1–3, USB |
|                  | <b>b</b>             | OFF, ON |                   |

| OAL Set Audio Output Level |                 |                  |                                |
|----------------------------|-----------------|------------------|--------------------------------|
| Transmitted command        | <b>OAL:a,b;</b> |                  |                                |
|                            | <b>a</b>        | V-160HD          | MAIN, AUX1–3, USB              |
|                            |                 | V-80HD           | MAIN, AUX1–2, USB, STREAM, REC |
|                            |                 | VR-120HD         | MAIN, AUX1–3, USB, STREAM, REC |
|                            |                 | VR-6HD           | MAIN, AUX1–2, USB, STREAM, REC |
|                            |                 | V-1-4K           | MAIN, AUX1–2, USB              |
|                            | <b>b</b>        | -INF, -800–0–100 | Level: -INF, -80.0–0.0–10.0dB  |
| Response command           | <b>ACK;</b>     |                  |                                |

| QOAL Get Audio Output Level |                     |                  |                                |
|-----------------------------|---------------------|------------------|--------------------------------|
| Transmitted command         | <b>QOAL:a;</b>      |                  |                                |
|                             | <b>a</b>            | V-160HD          | MAIN, AUX1–3, USB              |
|                             |                     | V-80HD           | MAIN, AUX1–2, USB, STREAM, REC |
|                             |                     | VR-120HD         | MAIN, AUX1–3, USB, STREAM, REC |
|                             |                     | VR-6HD           | MAIN, AUX1–2, USB, STREAM, REC |
|                             |                     | V-1-4K           | MAIN, AUX1–2, USB              |
|                             |                     |                  |                                |
| Response command            | <b>OAL:a,b;ACK;</b> |                  |                                |
|                             | <b>a</b>            | V-160HD          | MAIN, AUX1–3, USB              |
|                             |                     | V-80HD           | MAIN, AUX1–2, USB, STREAM, REC |
|                             |                     | VR-120HD         | MAIN, AUX1–3, USB, STREAM, REC |
|                             |                     | VR-6HD           | MAIN, AUX1–2, USB, STREAM, REC |
|                             |                     | V-1-4K           | MAIN, AUX1–2, USB              |
|                             | <b>b</b>            | -INF, -800–0–100 | Level: -INF, -80.0–0.0–10.0dB  |

| OAM Set Audio Output Mute ON/OFF |          |          |                                |
|----------------------------------|----------|----------|--------------------------------|
| Transmitted command              | OAM:a,b; |          |                                |
|                                  | a        | V-160HD  | MAIN, AUX1–3, USB              |
|                                  |          | V-80HD   | MAIN, AUX1–2, USB, STREAM, REC |
|                                  |          | VR-120HD | MAIN, AUX1–3, USB, STREAM, REC |
|                                  |          | VR-6HD   | MAIN, AUX1–2, USB, STREAM, REC |
|                                  |          | V-1-4K   | MAIN, AUX1–2, USB              |
|                                  | b        | OFF, ON  |                                |
| Response command                 | ACK;     |          |                                |

| QOAM Get status of Audio Output Mute |                |          |                                |
|--------------------------------------|----------------|----------|--------------------------------|
| Transmitted command                  | <b>QOAM:a;</b> |          |                                |
|                                      | <b>a</b>       | V-160HD  | MAIN, AUX1–3, USB              |
|                                      |                | V-80HD   | MAIN, AUX1–2, USB, STREAM, REC |
|                                      |                | VR-120HD | MAIN, AUX1–3, USB, STREAM, REC |
|                                      |                | VR-6HD   | MAIN, AUX1–2, USB, STREAM, REC |
|                                      |                | V-1-4K   | MAIN, AUX1–2, USB              |
|                                      |                |          |                                |

|                  |                     |          |                                |
|------------------|---------------------|----------|--------------------------------|
| Response command | <b>OAM:a,b;ACK;</b> |          |                                |
|                  | <b>a</b>            | V-160HD  | MAIN, AUX1–3, USB              |
|                  |                     | V-80HD   | MAIN, AUX1–2, USB, STREAM, REC |
|                  |                     | VR-120HD | MAIN, AUX1–3, USB, STREAM, REC |
|                  |                     | VR-6HD   | MAIN, AUX1–2, USB, STREAM, REC |
|                  |                     | V-1-4K   | MAIN, AUX1–2, USB              |
|                  | <b>b</b>            | OFF, ON  |                                |

|                     |                                                       |  |  |
|---------------------|-------------------------------------------------------|--|--|
| <b>RVB</b>          | Change Reverb ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |  |  |
| Transmitted command | <b>RVB;</b>                                           |  |  |
| Response command    | <b>ACK;</b>                                           |  |  |

|                     |        |                                                    |  |
|---------------------|--------|----------------------------------------------------|--|
| RVB                 |        | Set Reverb ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD) |  |
| Transmitted command | RVB:a; |                                                    |  |
|                     | a      | OFF, ON                                            |  |
| Response command    | ACK;   |                                                    |  |

|                     |                                                       |         |  |
|---------------------|-------------------------------------------------------|---------|--|
| <b>QRVB</b>         | Get status of Reverb (V-160HD/V-80HD/VR-120HD/VR-6HD) |         |  |
| Transmitted command | <b>QRVB;</b>                                          |         |  |
| Response command    | <b>RVB:a;ACK;</b>                                     |         |  |
|                     | <b>a</b>                                              | OFF, ON |  |

|                     |                                 |  |  |
|---------------------|---------------------------------|--|--|
| <b>ATM</b>          | Change Audio Auto Mixing ON/OFF |  |  |
| Transmitted command | <b>ATM;</b>                     |  |  |
| Response command    | <b>ACK;</b>                     |  |  |

|                     |                              |         |  |
|---------------------|------------------------------|---------|--|
| ATM                 | Set Audio Auto Mixing ON/OFF |         |  |
| Transmitted command | ATM:a;                       |         |  |
|                     | a                            | OFF, ON |  |
| Response command    | ACK;                         |         |  |

|                     |                                 |         |  |
|---------------------|---------------------------------|---------|--|
| <b>QATM</b>         | Get status of Audio Auto Mixing |         |  |
| Transmitted command | <b>QATM;</b>                    |         |  |
| Response command    | <b>ATM:a;ACK;</b>               |         |  |
|                     | <b>a</b>                        | OFF, ON |  |

## METER

| MTRLV Get level information for audio level meter |                             |          |                                                                                                                                                                                                     |
|---------------------------------------------------|-----------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transmitted command                               | <b>MTRLV:a;</b><br><b>a</b> |          |                                                                                                                                                                                                     |
|                                                   |                             |          | PFL, AFL                                                                                                                                                                                            |
| Response command                                  | <b>MTRLV:a0,a1,...;ACK;</b> |          |                                                                                                                                                                                                     |
|                                                   |                             |          | -INF, -80-0-10 Level: -INF, -80-0-10dB                                                                                                                                                              |
|                                                   | <b>a0-a53</b>               | V-160HD  | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-HDMI1-L-8-R, IN-SDI1-L-8-R, OUT-MAIN-L-R, OUT-AUX1-L-3-R, OUT-USB-L-R, OUT-DSK1-L-2-R                                                    |
|                                                   | <b>a0-a37</b>               | V-80HD   | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-4-R, IN-SDI1-L-4-R, IN-V.PLAYER-L-R, OUT-MAIN-L-R, OUT-AUX1-L-2-R, OUT-USB-L-R, OUT-STREAM-L-R                  |
|                                                   | <b>a0-a55</b>               | VR-120HD | IN-XLR1-6, IN-RCA1-L-2-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-SDI1-L-6-R, IN-V.PLAYER-L-R, OUT-MAIN-L-R, OUT-AUX1-L-3-R, OUT-USB-L-R, OUT-STREAM-L-R, OUT-MONITOR-L-R |
|                                                   | <b>a0-a39</b>               | VR-6HD   | IN-XLR1-6, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-V.PLAYER-L-R, OUT-MAIN-L-R, OUT-AUX1-L-2-R, OUT-USB-L-R, OUT-STREAM-L-R, OUT-MONITOR-L-R                |
|                                                   | <b>a0-a20</b>               | V-1-4K   | IN-XLR1-L-R, IN-USB-L-R, IN-HDMI1-L-5-R, OUT-MAIN-L-R, OUT-AUX1-L-2-R, OUT-USB-L-R                                                                                                                  |

## Examples of settings

|                     |                                   |                                                |
|---------------------|-----------------------------------|------------------------------------------------|
| Transmitted command | <b>MTRLV:PFL;</b>                 | Check PFL audio level meter                    |
| Response command    | <b>MTRLV:-INF,-80,...,10;ACK;</b> | Join level information with comma and transmit |

| MTRCH Get channel information for audio level meter |                             |          |                                                                                                                                                                                                     |
|-----------------------------------------------------|-----------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transmitted command                                 | <b>MTRCH:a;</b><br><b>a</b> |          |                                                                                                                                                                                                     |
|                                                     |                             | V-160HD  | 0-53                                                                                                                                                                                                |
|                                                     |                             | V-80HD   | 0-37                                                                                                                                                                                                |
|                                                     |                             | VR-120HD | 0-55                                                                                                                                                                                                |
|                                                     |                             | VR-6HD   | 0-39                                                                                                                                                                                                |
|                                                     |                             | V-1-4K   | 0-21                                                                                                                                                                                                |
| Response command                                    | <b>MTRCH:a,b;ACK;</b>       |          |                                                                                                                                                                                                     |
|                                                     | <b>a</b>                    | V-160HD  | 0-53                                                                                                                                                                                                |
|                                                     |                             | V-80HD   | 0-37                                                                                                                                                                                                |
|                                                     |                             | VR-120HD | 0-55                                                                                                                                                                                                |
|                                                     |                             | VR-6HD   | 0-39                                                                                                                                                                                                |
|                                                     |                             | V-1-4K   | 0-21                                                                                                                                                                                                |
|                                                     | <b>b</b>                    | V-160HD  | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-HDMI1-L-8-R, IN-SDI1-L-8-R, OUT-MAIN-L-R, OUT-AUX1-L-3-R, OUT-USB-L-R, OUT-DSK1-L-2-R                                                    |
|                                                     |                             | V-80HD   | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-4-R, IN-SDI1-L-4-R, IN-V.PLAYER-L-R, OUT-MAIN-L-R, OUT-AUX1-L-2-R, OUT-USB-L-R, OUT-STREAM-L-R                  |
|                                                     |                             | VR-120HD | IN-XLR1-6, IN-RCA1-L-2-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-SDI1-L-6-R, IN-V.PLAYER-L-R, OUT-MAIN-L-R, OUT-AUX1-L-3-R, OUT-USB-L-R, OUT-STREAM-L-R, OUT-MONITOR-L-R |
|                                                     |                             | VR-6HD   | IN-XLR1-6, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-V.PLAYER-L-R, OUT-MAIN-L-R, OUT-AUX1-L-2-R, OUT-USB-L-R, OUT-STREAM-L-R, OUT-MONITOR-L-R                |
|                                                     |                             | V-1-4K   | IN-XLR1-L-R, IN-USB-L-R, IN-HDMI1-L-5-R, OUT-MAIN-L-R, OUT-AUX1-L-2-R, OUT-USB-L-R                                                                                                                  |

## Examples of settings

|                     |                             |                                |
|---------------------|-----------------------------|--------------------------------|
| Transmitted command | <b>MTRCH:0;</b>             | Get zeroth channel information |
| Response command    | <b>MTRCH:0,IN-XLR1;ACK;</b> | Zeroth is IN-XLR1              |

| GRLV Get level information for Comp GR level |                            |                |                                                                                                                   |
|----------------------------------------------|----------------------------|----------------|-------------------------------------------------------------------------------------------------------------------|
| Transmitted command                          | <b>GRLV;</b>               |                |                                                                                                                   |
| Response command                             | <b>GRLV:a0,a1,...;ACK;</b> | -INF, -80-0-10 | Level: -INF, -80-0-10dB                                                                                           |
|                                              | <b>a0-a23</b>              | V-160HD        | IN-XLR1-2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-HDMI1-8, IN-SDI1-8, OUT-MAIN-LO, OUT-MAIN-MID, OUT-MAIN-HIGH         |
|                                              | <b>a0-a17</b>              | V-80HD         | IN-XLR1-2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-4, IN-SDI1-4, IN-V.PLAYER, OUT-MAIN, OUT-AUX1-2   |
|                                              | <b>a0-a29</b>              | VR-120HD       | IN-XLR1-6, IN-RCA1-2, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-6, IN-SDI1-6, IN-V.PLAYER, OUT-MAIN, OUT-AUX1-3 |
|                                              | <b>a0-a19</b>              | VR-6HD         | IN-XLR1-6, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-6, IN-V.PLAYER, OUT-MAIN, OUT-AUX1-2              |
|                                              | <b>a0-a9</b>               | V-1-4K         | IN-XLR1-L-R, IN-USB, IN-HDMI1-5, OUT-MAIN, OUT-AUX1-2                                                             |

| GRCH Get channel information for Comp GR level |                  |          |                                                                                                                   |
|------------------------------------------------|------------------|----------|-------------------------------------------------------------------------------------------------------------------|
| Transmitted command                            | <b>GRCH:a;</b>   |          |                                                                                                                   |
| Response command                               | <b>a</b>         | V-160HD  | 0-23                                                                                                              |
|                                                |                  | V-80HD   | 0-17                                                                                                              |
|                                                |                  | VR-120HD | 0-27                                                                                                              |
|                                                |                  | VR-6HD   | 0-19                                                                                                              |
|                                                |                  | V-1-4K   | 0-9                                                                                                               |
|                                                |                  |          |                                                                                                                   |
| Response command                               | <b>GRCH:a,b;</b> |          |                                                                                                                   |
|                                                | <b>a</b>         | V-160HD  | 0-23                                                                                                              |
|                                                |                  | V-80HD   | 0-17                                                                                                              |
|                                                |                  | VR-120HD | 0-27                                                                                                              |
|                                                |                  | VR-6HD   | 0-19                                                                                                              |
|                                                |                  | V-1-4K   | 0-9                                                                                                               |
|                                                | <b>b</b>         | V-160HD  | IN-XLR1-2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-HDMI1-8, IN-SDI1-8, OUT-MAIN-LO, OUT-MAIN-MID, OUT-MAIN-HIGH         |
|                                                |                  | V-80HD   | IN-XLR1-2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-4, IN-SDI1-4, IN-V.PLAYER, OUT-MAIN, OUT-AUX1-2   |
|                                                |                  | VR-120HD | IN-XLR1-6, IN-RCA1-2, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-6, IN-SDI1-6, IN-V.PLAYER, OUT-MAIN, OUT-AUX1-3 |
|                                                |                  | VR-6HD   | IN-XLR1-6, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-6, IN-V.PLAYER, OUT-MAIN, OUT-AUX1-2              |
|                                                |                  | V-1-4K   | IN-XLR1, IN-USB, IN-HDMI1-5, OUT-MAIN, OUT-AUX1-2                                                                 |

| AMLV Get level information for Auto Mixing level |                            |                |                                                                                             |
|--------------------------------------------------|----------------------------|----------------|---------------------------------------------------------------------------------------------|
| Transmitted command                              | <b>AMLV;</b>               |                |                                                                                             |
| Response command                                 | <b>AMLV:a0,a1,...;ACK;</b> | -INF, -80-0-10 | Level: -INF, -80-0-10dB                                                                     |
|                                                  | <b>a0-a20</b>              | V-160HD        | IN-XLR1-2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-HDMI1-8, IN-SDI1-8                             |
|                                                  | <b>a0-a14</b>              | V-80HD         | IN-XLR1-2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-4, IN-SDI1-4, IN-V.PLAYER   |
|                                                  | <b>a0-a23</b>              | VR-120HD       | IN-XLR1-6, IN-RCA1-2, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-6, IN-SDI1-6, IN-V.PLAYER |
|                                                  | <b>a0-a16</b>              | VR-6HD         | IN-XLR1-6, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1-6, IN-V.PLAYER              |
|                                                  | <b>a0-a6</b>               | V-1-4K         | IN-XLR1, IN-USB, IN-HDMI1-5                                                                 |

| AMCH Get channel information for Auto Mixing level |                |          |      |
|----------------------------------------------------|----------------|----------|------|
| Transmitted command                                | <b>AMCH:a;</b> |          |      |
| Response command                                   | <b>a</b>       | V-160HD  | 0-20 |
|                                                    |                | V-80HD   | 0-14 |
|                                                    |                | VR-120HD | 0-23 |
|                                                    |                | VR-6HD   | 0-16 |
|                                                    |                | V-1-4K   | 0-6  |
|                                                    |                |          |      |

## Basic control commands

| Response command | <b>AMCH:a,b;</b> |          |                                                                                             |
|------------------|------------------|----------|---------------------------------------------------------------------------------------------|
|                  | <b>a</b>         | V-160HD  | 0–20                                                                                        |
|                  |                  | V-80HD   | 0–14                                                                                        |
|                  |                  | VR-120HD | 0–23                                                                                        |
|                  |                  | VR-6HD   | 0–16                                                                                        |
|                  |                  | V-1-4K   | 0–6                                                                                         |
|                  | <b>b</b>         | V-160HD  | IN-XLR1–2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-HDMI1–8, IN-SDI1–8                             |
|                  |                  | V-80HD   | IN-XLR1–2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1–4, IN-SDI1–4, IN-V.PLAYER   |
|                  |                  | VR-120HD | IN-XLR1–6, IN-RCA1–2, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1–6, IN-SDI1–6, IN-V.PLAYER |
|                  |                  | VR-6HD   | IN-XLR1–6, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1–6, IN-V.PLAYER              |
|                  |                  | V-1-4K   | IN-XLR1, IN-USB, IN-HDMI1–5                                                                 |

| SPLV                |                     | Get level information for Sig/Peak level |                |
|---------------------|---------------------|------------------------------------------|----------------|
| Transmitted command | SPLV;               |                                          |                |
| Response command    | SPLV:a0,a1,...;ACK; |                                          | -INF, -80-0-10 |

| SPCH                |           | Get channel information for Sig/Peak level |                                                                                                                                                     |
|---------------------|-----------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Transmitted command | SPCH:a;   |                                            |                                                                                                                                                     |
|                     | a         | V-160HD                                    | 0–27                                                                                                                                                |
|                     |           | V-80HD                                     | 0–19                                                                                                                                                |
|                     |           | VR-120HD                                   | 0–30                                                                                                                                                |
|                     |           | VR-6HD                                     | 0–22                                                                                                                                                |
|                     |           | V-1-4K                                     | 0–10                                                                                                                                                |
| Response command    | SPCH:a,b; |                                            |                                                                                                                                                     |
|                     | a         | V-160HD                                    | 0–27                                                                                                                                                |
|                     |           | V-80HD                                     | 0–19                                                                                                                                                |
|                     |           | VR-120HD                                   | 0–30                                                                                                                                                |
|                     |           | VR-6HD                                     | 0–22                                                                                                                                                |
|                     |           | V-1-4K                                     | 0–10                                                                                                                                                |
|                     | b         | V-160HD                                    | IN-XLR1–2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-HDMI1–8, IN-SDI1–8, OUT-MAIN, OUT-AUX1–3, OUT-USB, OUT-DSK1–2                                          |
|                     |           | V-80HD                                     | IN-XLR1–2, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1–4, IN-SDI1–4, IN-V.PLAYER, OUT-MAIN, OUT-AUX1–2, OUT-USB, OUT-STRAM                 |
|                     |           | VR-120HD                                   | IN-XLR1–6, IN-RCA1–2, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1–6, IN-SDI1–6, IN-V.PLAYER, OUT-MAIN, OUT-AUX1–3, OUT-USB, OUT-STREAM, OUT-MONITOR |
|                     |           | VR-6HD                                     | IN-XLR1–6, IN-RCA1, IN-USB, IN-BLUETOOTH, IN-A.PLAYER, IN-HDMI1–6, IN-V.PLAYER, OUT-MAIN, OUT-AUX1–2, OUT-USB, OUT-STREAM, OUT-MONITOR              |
|                     |           | V-1-4K                                     | IN-XLR1, IN-USB, IN-HDMI1–5, OUT-MAIN, OUT-AUX1–2, OUT-USB                                                                                          |
|                     |           |                                            |                                                                                                                                                     |

| AUXLV               |                       | Get information on AUX level |                                                                                                                         |
|---------------------|-----------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Transmitted command | AUXLV:a;              |                              |                                                                                                                         |
|                     | a                     | V-160HD                      | AUX1-3                                                                                                                  |
|                     |                       | V-80HD                       | AUX1-2                                                                                                                  |
|                     |                       | VR-120HD                     | AUX1-3                                                                                                                  |
|                     |                       | VR-6HD                       | AUX1-2                                                                                                                  |
|                     |                       | V-1-4K                       | AUX1-2                                                                                                                  |
| Response command    | QAUXLV:a0,a1,...;ACK; |                              | -INF, -80-0-10                      Level: -INF, -80-0-10dB                                                             |
|                     | a0-a39                | V-160HD                      | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-HDMI1-L-8-R, IN-SDI1-L-8-R                                   |
|                     | a0-a27                | V-80HD                       | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-4-R, IN-SDI1-L-4-R, IN-V.PLAYER-L-R |
|                     | a0-a41                | VR-120HD                     | IN-XLR1-6, IN-RCA1-L-2-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-SDI1-L-6-R, IN-V.PLAYER-L-R |
|                     | a0-a27                | VR-6HD                       | IN-XLR1-6, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-V.PLAYER-L-R                |
|                     | a0-a21                | V-1-4K                       | IN-XLR1-L-R, IN-USB-L-R, IN-HDMI1-L-5-R                                                                                 |
|                     |                       |                              |                                                                                                                         |
| AUXCH               |                       | Get AUX channel information  |                                                                                                                         |
| Transmitted command | AUXCH:a;              |                              |                                                                                                                         |
|                     | a                     | V-160HD                      | 0-39                                                                                                                    |
|                     |                       | V-80HD                       | 0-27                                                                                                                    |
|                     |                       | VR-120HD                     | 0-41                                                                                                                    |
|                     |                       | VR-6HD                       | 0-27                                                                                                                    |
|                     |                       | V-1-4K                       | 0-13                                                                                                                    |
| Response command    | QAUXCH:a,b;           |                              |                                                                                                                         |
|                     | a                     | V-160HD                      | 0-39                                                                                                                    |
|                     |                       | V-80HD                       | 0-27                                                                                                                    |
|                     |                       | VR-120HD                     | 0-41                                                                                                                    |
|                     |                       | VR-6HD                       | 0-27                                                                                                                    |
|                     |                       | V-1-4K                       | 0-13                                                                                                                    |
|                     | b                     | V-160HD                      | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-HDMI1-L-8-R, IN-SDI1-L-8-R                                   |
|                     |                       | V-80HD                       | IN-XLR1-2, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-4-R, IN-SDI1-L-4-R, IN-V.PLAYER-L-R |
|                     |                       | VR-120HD                     | IN-XLR1-6, IN-RCA1-L-2-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-SDI1-L-6-R, IN-V.PLAYER-L-R |
|                     |                       | VR-6HD                       | IN-XLR1-6, IN-RCA1-L-1-R, IN-USB-L-R, IN-BLUETOOTH-L-R, IN-A.PLAYER-L-R, IN-HDMI1-L-6-R, IN-V.PLAYER-L-R                |
|                     |                       | V-1-4K                       | IN-XLR1-L-R, IN-USB-L-R, IN-HDMI1-L-5-R                                                                                 |
|                     |                       |                              |                                                                                                                         |

## CONTROL

| MEM Recall scene memory |               |          |            |
|-------------------------|---------------|----------|------------|
| Transmitted command     | <b>MEM:a;</b> |          |            |
|                         | <b>a</b>      | V-160HD  | MEMORY1–30 |
|                         |               | V-80HD   | MEMORY1–32 |
|                         |               | VR-120HD | MEMORY1–32 |
|                         |               | VR-6HD   | MEMORY1–32 |
|                         |               | V-1-4K   | MEMORY1–8  |
| Response command        | <b>ACK;</b>   |          |            |

| QMEM Get selected scene memory |                   |          |                 |
|--------------------------------|-------------------|----------|-----------------|
| Transmitted command            | <b>QMEM;</b>      |          |                 |
| Response command               | <b>MEM:a;ACK;</b> |          |                 |
|                                | <b>a</b>          | V-160HD  | N/A, MEMORY1–30 |
|                                |                   | V-80HD   | N/A, MEMORY1–32 |
|                                |                   | VR-120HD | N/A, MEMORY1–32 |
|                                |                   | VR-6HD   | N/A, MEMORY1–32 |
|                                |                   | V-1-4K   | N/A, MEMORY1–8  |

| GPO                 |        | Output GPO (ONE SHOT) (V-160HD/V-80HD/VR-120HD/VR-6HD) |  |
|---------------------|--------|--------------------------------------------------------|--|
| Transmitted command | GPO:a; |                                                        |  |
|                     | a      | GPO1–16                                                |  |
| Response command    | ACK;   |                                                        |  |

| GPO                 |          | Output GPO (ALTERNATE) (V-160HD/V-80HD/VR-120HD/VR-6HD) |  |
|---------------------|----------|---------------------------------------------------------|--|
| Transmitted command | GPO:a,b; |                                                         |  |
|                     | a        | GPO1–16                                                 |  |
|                     | b        | OFF, ON                                                 |  |
| Response command    | ACK;     |                                                         |  |

| QGPO                |              |         | Get status of GPO (V-160HD/V-80HD/VR-120HD/VR-6HD) |
|---------------------|--------------|---------|----------------------------------------------------|
| Transmitted command | QGPO:a;      |         |                                                    |
|                     | a            | GPO1-16 |                                                    |
| Response command    | GPO:a,b;ACK; |         |                                                    |
|                     | a            | GPO1-16 |                                                    |
|                     | b            | OFF, ON |                                                    |

| TLY Get status of tally |             |  |  |
|-------------------------|-------------|--|--|
| Transmitted command     | <b>TLY;</b> |  |  |



|                  |                           |          |                                                        |                                      |
|------------------|---------------------------|----------|--------------------------------------------------------|--------------------------------------|
| Response command | <b>TLY:a0,a1,...;ACK;</b> |          | 0–3                                                    | 0: OFF, 1: PGM, 2: PST, 3: PGM & PST |
|                  | <b>a0–a51</b>             | V-160HD  | HDMI1–8, SDI1–8, STILL1–16, INPUT1–20                  |                                      |
|                  | <b>a0–a56</b>             | V-80HD   | HDMI1–4, SDI1–4, STILL1–32, V.PLAYER or SRT, INPUT1–16 |                                      |
|                  | <b>a0–a36</b>             | VR-120HD | HDMI1–6, SDI1–6, STILL1–16, V.PLAYER or SRT, INPUT1–8  |                                      |
|                  | <b>a0–a28</b>             | VR-6HD   | HDMI1–6, STILL1–16, V.PLAYER or SRT, INPUT1–6          |                                      |
|                  | <b>a0–a14</b>             | V-1-4K   | HDMI1–5, STILL1–2, INPUT1–8                            |                                      |

### Examples of settings

|                     |                              |                                                                     |
|---------------------|------------------------------|---------------------------------------------------------------------|
| Transmitted command | <b>TLY;</b>                  | Get status of tally                                                 |
| Response command    | <b>TLY:0,1,2,0...,0;ACK;</b> | Express tally information as numbers, join with commas and transmit |

### ASW Switch between Auto Switching ON/OFF

|                     |             |
|---------------------|-------------|
| Transmitted command | <b>ASW;</b> |
| Response command    | <b>ACK;</b> |

### ASW Set Auto Switching ON/OFF

|                     |               |         |
|---------------------|---------------|---------|
| Transmitted command | <b>ASW:a;</b> |         |
|                     | <b>a</b>      | OFF, ON |
| Response command    | <b>ACK;</b>   |         |

### QASW Get status of Auto Switching

|                     |            |         |  |
|---------------------|------------|---------|--|
| Transmitted command | QASW;      |         |  |
| Response command    | ASW:a;ACK; |         |  |
|                     | a          | OFF, ON |  |

### INSC Execute Input Scan

|                     |                |                         |
|---------------------|----------------|-------------------------|
| Transmitted command | <b>INSC:a;</b> |                         |
|                     | <b>a</b>       | NORMAL, REVERSE, RANDOM |
| Response command    | <b>ACK;</b>    |                         |

### MEMSC Execute Scene Memory Scan (V-160HD/V-80HD/VR-120HD/VR-6HD)

|                     |                 |                         |
|---------------------|-----------------|-------------------------|
| Transmitted command | <b>MEMSC:a;</b> |                         |
|                     | <b>a</b>        | NORMAL, REVERSE, RANDOM |
| Response command    | <b>ACK;</b>     |                         |

### PPSC Execute PinP Source Scan (V-160HD/V-80HD/VR-120HD/VR-6HD)

|                     |                  |                         |         |
|---------------------|------------------|-------------------------|---------|
| Transmitted command | <b>PPSC:a,b;</b> |                         |         |
|                     | <b>a</b>         | V-160HD                 | PinP1–4 |
|                     |                  | V-80HD                  | PinP1–2 |
|                     |                  | VR-120HD                | PinP1–4 |
|                     |                  | VR-6HD                  | PinP1–2 |
|                     | <b>b</b>         | NORMAL, REVERSE, RANDOM |         |

## Basic control commands

|                                                                                        |                         |                                 |                    |
|----------------------------------------------------------------------------------------|-------------------------|---------------------------------|--------------------|
| Response command                                                                       | ACK;                    |                                 |                    |
|                                                                                        |                         |                                 |                    |
| DSKSC      Execute DSK Source Scan (V-160HD/V-80HD/VR-120HD/VR-6HD)                    |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | DSKSC:a,b;              |                                 |                    |
|                                                                                        | a                       | V-160HD                         | DSK1-2             |
|                                                                                        |                         | V-80HD                          | DSK1               |
|                                                                                        |                         | VR-120HD                        | DSK1-2             |
|                                                                                        |                         | VR-6HD                          | DSK1               |
| b                                                                                      | NORMAL, REVERSE, RANDOM |                                 |                    |
| Response command                                                                       | ACK;                    |                                 |                    |
|                                                                                        |                         |                                 |                    |
| STRTY      Set Stream & Record Type (V-80HD/VR-120HD/VR-6HD)                           |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | STRTY:a;                |                                 |                    |
|                                                                                        | a                       | RTMP, SRT                       |                    |
| Response command                                                                       | ACK;                    |                                 |                    |
|                                                                                        |                         |                                 |                    |
| QSTRTY      Get Stream & Record Type (V-80HD/VR-120HD/VR-6HD)                          |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | QSTRTY;                 |                                 |                    |
| Response command                                                                       | STRTY:a;ACK;            |                                 |                    |
|                                                                                        | a                       | RTMP, SRT                       |                    |
|                                                                                        |                         |                                 |                    |
| STROA      Start/stop Stream & Record (V-80HD/VR-120HD/VR-6HD, enabled when Type=RTMP) |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | STROA:a;                |                                 |                    |
|                                                                                        | a                       | STOP, START                     |                    |
| Response command                                                                       | ACK;                    |                                 |                    |
|                                                                                        |                         |                                 |                    |
| SRTOSS      Start/stop SRT Out (V-80HD/VR-120HD/VR-6HD, enabled when Type=SRT)         |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | SRTOSS:a;               |                                 |                    |
|                                                                                        | a                       | STOP, START                     |                    |
| Response command                                                                       | ACK;                    |                                 |                    |
|                                                                                        |                         |                                 |                    |
| QSTRST      Get status of Stream & Record (V-80HD/VR-120HD/VR-6HD)                     |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | QSTRST;                 |                                 |                    |
| Response command                                                                       | STRST:a;ACK;            |                                 |                    |
|                                                                                        | a                       | STOP, STOPPING, STARTING, ONAIR |                    |
|                                                                                        |                         |                                 |                    |
| QSTRTM      Get elapsed time of Stream & Record (V-80HD/VR-120HD/VR-6HD)               |                         |                                 |                    |
|                                                                                        |                         |                                 |                    |
| Transmitted command                                                                    | QSTRTM;                 |                                 |                    |
| Response command                                                                       | STRTM:a;ACK;            |                                 |                    |
|                                                                                        | a                       | 0-                              | Elapsed time (sec) |

## Examples of settings

|                     |                      |                                        |
|---------------------|----------------------|----------------------------------------|
| Transmitted command | <b>QSTRTM;</b>       | Check the Stream & Record elapsed time |
| Response command    | <b>STRTM:30;ACK;</b> | Stream & Record elapsed time is 30 sec |

|                                                                                        |                |         |
|----------------------------------------------------------------------------------------|----------------|---------|
| <b>SFIM</b> Set Safety Image ON/OFF (V-80HD/VR-120HD/VR-6HD, enabled only when on air) |                |         |
| Transmitted command                                                                    | <b>SFIM:a;</b> |         |
|                                                                                        | <b>a</b>       | OFF, ON |
| Response command                                                                       | <b>ACK;</b>    |         |

|                                                                                            |                    |                  |
|--------------------------------------------------------------------------------------------|--------------------|------------------|
| <b>QSFIM</b> Get status of Safety Image (V-80HD/VR-120HD/VR-6HD, enabled only when on air) |                    |                  |
| Transmitted command                                                                        | <b>QSFIM;</b>      |                  |
| Response command                                                                           | <b>SFIM:a;ACK;</b> |                  |
|                                                                                            | <b>a</b>           | OFF, ON, DELAYED |

|                                                                      |                |         |
|----------------------------------------------------------------------|----------------|---------|
| <b>EXRC</b> Set External Rec Control ON/OFF (V-80HD/VR-120HD/VR-6HD) |                |         |
| Transmitted command                                                  | <b>EXRC:a;</b> |         |
|                                                                      | <b>a</b>       | OFF, ON |
| Response command                                                     | <b>ACK;</b>    |         |

|                                                                          |                    |         |
|--------------------------------------------------------------------------|--------------------|---------|
| <b>QEXRC</b> Get status of External Rec Control (V-80HD/VR-120HD/VR-6HD) |                    |         |
| Transmitted command                                                      | <b>QEXRC;</b>      |         |
| Response command                                                         | <b>EXRC:a;ACK;</b> |         |
|                                                                          | <b>a</b>           | OFF, ON |

|                                                            |                 |                                |
|------------------------------------------------------------|-----------------|--------------------------------|
| <b>APL</b> Play/stop Audio Player (V-80HD/VR-120HD/VR-6HD) |                 |                                |
| Transmitted command                                        | <b>APL:a,b;</b> |                                |
|                                                            | <b>a</b>        | V-80HD A-1 – A-8 – H-1 – H-8   |
|                                                            |                 | VR-120HD A-1 – A-8 – B-1 – B-8 |
|                                                            |                 | VR-6HD A-1 – A-6 – B-1 – B-6   |
|                                                            | <b>b</b>        | STOP, PLAY, PAUSE, RESUME      |
| Response command                                           | <b>ACK;</b>     |                                |

|                                                                          |                     |                                    |
|--------------------------------------------------------------------------|---------------------|------------------------------------|
| <b>QAPS</b> Get playback status of Audio Player (V-80HD/VR-120HD/VR-6HD) |                     |                                    |
| Transmitted command                                                      | <b>QAPS:a;</b>      |                                    |
|                                                                          | <b>a</b>            | V-80HD A-1 – A-8 – H-1 – H-8       |
|                                                                          |                     | VR-120HD A-1 – A-8 – B-1 – B-8     |
|                                                                          |                     | VR-6HD A-1 – A-6 – B-1 – B-6       |
| Response command                                                         | <b>APS:a,b;ACK;</b> |                                    |
|                                                                          | <b>a</b>            | V-80HD A-1 – A-8 – H-1 – H-8       |
|                                                                          |                     | VR-120HD A-1 – A-8 – B-1 – B-8     |
|                                                                          |                     | VR-6HD A-1 – A-6 – B-1 – B-6       |
|                                                                          | <b>b</b>            | STOP, PLAY, PAUSE, FADEIN, FADEOUT |

## Basic control commands

| QAPE Get import status of Audio Player (V-80HD/VR-120HD/VR-6HD) |                     |          |                       |
|-----------------------------------------------------------------|---------------------|----------|-----------------------|
| Transmitted command                                             | <b>QAPE:a;</b>      |          |                       |
|                                                                 | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                                                                 |                     | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                                                                 |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6 |
| Response command                                                | <b>APE:a,b;ACK;</b> |          |                       |
|                                                                 | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                                                                 |                     | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                                                                 |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6 |
|                                                                 | <b>b</b>            | OFF, ON  |                       |

| QAPD Get length of Audio Player file (V-80HD/VR-120HD/VR-6HD) |                     |          |                       |
|---------------------------------------------------------------|---------------------|----------|-----------------------|
| Transmitted command                                           | <b>QAPD:a;</b>      |          |                       |
|                                                               | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                                                               |                     | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                                                               |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6 |
| Response command                                              | <b>APD:a,b;ACK;</b> |          |                       |
|                                                               | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                                                               |                     | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                                                               |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6 |
|                                                               | <b>b</b>            | 0–       | File length (msec)    |

| QAPC Get current playback time of Audio Player (V-80HD/VR-120HD/VR-6HD) |                     |          |                              |
|-------------------------------------------------------------------------|---------------------|----------|------------------------------|
| Transmitted command                                                     | <b>QAPC:a;</b>      |          |                              |
|                                                                         | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8        |
|                                                                         |                     | VR-120HD | A-1 – A-8 – B-1 – B-8        |
|                                                                         |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6        |
| Response command                                                        | <b>APC:a,b;ACK;</b> |          |                              |
|                                                                         | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8        |
|                                                                         |                     | VR-120HD | A-1 – A-8 – B-1 – B-8        |
|                                                                         |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6        |
|                                                                         | <b>b</b>            | 0–       | Current playback time (msec) |

| APR Set repeat for Audio Player (V-80HD/VR-120HD/VR-6HD) |          |          |                       |
|----------------------------------------------------------|----------|----------|-----------------------|
| Transmitted command                                      | APR:a,b; |          |                       |
|                                                          | a        | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                                                          |          | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                                                          |          | VR-6HD   | A-1 – A-6 – B-1 – B-6 |
|                                                          | b        | OFF, ON  |                       |
| Response command                                         | ACK;     |          |                       |

| QAPR Get repeat status for Audio Player (V-80HD/VR-120HD/VR-6HD) |                |          |                       |
|------------------------------------------------------------------|----------------|----------|-----------------------|
| Transmitted command                                              | <b>QAPR:a;</b> |          |                       |
|                                                                  | <b>a</b>       | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                                                                  |                | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                                                                  |                | VR-6HD   | A-1 – A-6 – B-1 – B-6 |

|                  |                     |          |                       |
|------------------|---------------------|----------|-----------------------|
| Response command | <b>APR:a,b;ACK;</b> |          |                       |
|                  | <b>a</b>            | V-80HD   | A-1 – A-8 – H-1 – H-8 |
|                  |                     | VR-120HD | A-1 – A-8 – B-1 – B-8 |
|                  |                     | VR-6HD   | A-1 – A-6 – B-1 – B-6 |
|                  | <b>b</b>            | OFF, ON  |                       |

| PBTY Set Video Player/SRT In Type (V-80HD/VR-120HD/VR-6HD) |                |
|------------------------------------------------------------|----------------|
| Transmitted command                                        | PBTY:a;        |
|                                                            | aV.PLAYER, SRT |
| Response command                                           | ACK;           |

|                                                                    |                    |               |  |
|--------------------------------------------------------------------|--------------------|---------------|--|
| <b>QPBTY</b> Get Video Player/SRT In Type (V-80HD/VR-120HD/VR-6HD) |                    |               |  |
| Transmitted command                                                | <b>QPBTY;</b>      |               |  |
| Response command                                                   | <b>PBTY:a;ACK;</b> |               |  |
|                                                                    | <b>a</b>           | V.PLAYER, SRT |  |

|                     |        |                                                                              |  |
|---------------------|--------|------------------------------------------------------------------------------|--|
| VPL                 |        | Start/stop Video Player (V-80HD/VR-120HD/VR-6HD, enabled when Type=V.PLAYER) |  |
| Transmitted command | VPL:a; |                                                                              |  |
|                     | a      | STOP, PLAY, PAUSE, RESUME                                                    |  |
| Response command    | ACK;   |                                                                              |  |

|                                                                                             |                   |                                       |  |
|---------------------------------------------------------------------------------------------|-------------------|---------------------------------------|--|
| <b>QVPS</b> Get status of Video Player (V-80HD/VR-120HD/VR-6HD, enabled when Type=V.PLAYER) |                   |                                       |  |
| Transmitted command                                                                         | <b>QVPS;</b>      |                                       |  |
| Response command                                                                            | <b>VPS:a;ACK;</b> |                                       |  |
|                                                                                             | <b>a</b>          | STOP, PLAY, PAUSE, STARTING, STOPPING |  |

|                                                                                                    |                   |                            |  |
|----------------------------------------------------------------------------------------------------|-------------------|----------------------------|--|
| <b>QVPE</b> Get import status of Video Player (V-80HD/VR-120HD/VR-6HD, enabled when Type=V.PLAYER) |                   |                            |  |
| Transmitted command                                                                                | <b>QVPE;</b>      |                            |  |
| Response command                                                                                   | <b>VPE:a;ACK;</b> |                            |  |
|                                                                                                    | <b>a</b>          | N/A, INTERNAL, SDCARD, USB |  |

|                                                                                                  |                   |    |                    |
|--------------------------------------------------------------------------------------------------|-------------------|----|--------------------|
| <b>QVPD</b> Get file length of Video Player (V-80HD/VR-120HD/VR-6HD, enabled when Type=V.PLAYER) |                   |    |                    |
| Transmitted command                                                                              | <b>QVPD;</b>      |    |                    |
| Response command                                                                                 | <b>VPD:a;ACK;</b> |    |                    |
|                                                                                                  | <b>a</b>          | 0– | File length (msec) |

|                                                                                                            |                   |    |                              |
|------------------------------------------------------------------------------------------------------------|-------------------|----|------------------------------|
| <b>QVPC</b> Get current playback time of Video Player (V-80HD/VR-120HD/VR-6HD, enabled when Type=V.PLAYER) |                   |    |                              |
| Transmitted command                                                                                        | <b>QVPC;</b>      |    |                              |
| Response command                                                                                           | <b>VPC:a;ACK;</b> |    |                              |
|                                                                                                            | <b>a</b>          | 0– | Current playback time (msec) |

## Basic control commands

|                                                                                                     |                                                                                                                              |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>SRTISS</b> Start/stop SRT In (V-80HD/VR-120HD/VR-6HD, enabled when Type=SRT)                     |                                                                                                                              |
| Transmitted command                                                                                 | <b>SRTISS:a;</b><br><b>a</b> STOP, START                                                                                     |
| Response command                                                                                    | <b>ACK;</b>                                                                                                                  |
| <b>QSRTIST</b> Get SRT In status (V-80HD/VR-120HD/VR-6HD, enabled when Type=SRT)                    |                                                                                                                              |
| Transmitted command                                                                                 | <b>QSRTIST;</b>                                                                                                              |
| Response command                                                                                    | <b>SRTIST:a;</b><br><b>a</b> STOPPED, STOPPING, STARTING, STARTED                                                            |
| <b>STRSTB</b> Get whether Stream & Record and SRT Out can be set to ON AIR (V-80HD/VR-120HD/VR-6HD) |                                                                                                                              |
| Transmitted command                                                                                 | <b>STRSTB;</b>                                                                                                               |
| Response command                                                                                    | <b>STRSTB:a;ACK;</b><br><b>a</b> OK, ERROR_HDCP, ERROR_NO_SDCARD, ERROR_WRITE_PROTECTED, ERROR_SDXC_REQUIRED, ERROR_RESOURCE |
| <b>VPSTB</b> Get whether Video Player can play back (V-80HD/VR-120HD/VR-6HD)                        |                                                                                                                              |
| Transmitted command                                                                                 | <b>VPSTB;</b>                                                                                                                |
| Response command                                                                                    | <b>VPSTB:a;ACK;</b><br><b>a</b> OK, ERROR_RESOURCE, ERROR_NOT_FOUND                                                          |
| <b>SRTISTB</b> Get whether SRT In can start (V-80HD/VR-120HD/VR-6HD)                                |                                                                                                                              |
| Transmitted command                                                                                 | <b>SRTISTB;</b>                                                                                                              |
| Response command                                                                                    | <b>SRTISTB:a;ACK;</b><br><b>a</b> OK, ERROR_RESOURCE                                                                         |

## CAMERA

| CAMPT Operate Pan and Tilt for the PTZ camera |                   |          |            |
|-----------------------------------------------|-------------------|----------|------------|
| Transmitted command                           | CAMPT:a,b,c;      |          |            |
|                                               | a                 | V-160HD  | CAMERA1-16 |
|                                               |                   | V-80HD   | CAMERA1-8  |
|                                               |                   | VR-120HD | CAMERA1-12 |
|                                               |                   | VR-6HD   | CAMERA1-6  |
|                                               |                   | V-1-4K   | CAMERA1-5  |
| b                                             | LEFT, STOP, RIGHT |          |            |
| c                                             | DOWN, STOP, UP    |          |            |
| Response command                              | ACK;              |          |            |

| CAMPTS Set the pan and tilt speed for the PTZ camera |             |          |            |
|------------------------------------------------------|-------------|----------|------------|
| Transmitted command                                  | CAMPTS:a,b; |          |            |
|                                                      | a           | V-160HD  | CAMERA1-16 |
|                                                      |             | V-80HD   | CAMERA1-8  |
|                                                      |             | VR-120HD | CAMERA1-12 |
|                                                      |             | VR-6HD   | CAMERA1-6  |
|                                                      |             | V-1-4K   | CAMERA1-5  |
|                                                      | b           | 1-24     |            |
| Response command                                     | ACK;        |          |            |

| QCAMPTS Get the pan and tilt speed for the PTZ camera |               |          |            |
|-------------------------------------------------------|---------------|----------|------------|
| Transmitted command                                   | QCAMPTS:a;    |          |            |
|                                                       | a             | V-160HD  | CAMERA1-16 |
|                                                       |               | V-80HD   | CAMERA1-8  |
|                                                       |               | VR-120HD | CAMERA1-12 |
|                                                       |               | VR-6HD   | CAMERA1-6  |
|                                                       |               | V-1-4K   | CAMERA1-5  |
| Response command                                      | CPTS:a,b;ACK; |          |            |
|                                                       | a             | V-160HD  | CAMERA1-16 |
|                                                       |               | V-80HD   | CAMERA1-8  |
|                                                       |               | VR-120HD | CAMERA1-12 |
|                                                       |               | VR-6HD   | CAMERA1-6  |
|                                                       |               | V-1-4K   | CAMERA1-5  |
|                                                       | b             | 1-24     |            |

| CAMZM Operate the Zoom for the PTZ camera |                                                  |          |            |
|-------------------------------------------|--------------------------------------------------|----------|------------|
| Transmitted command                       | CAMZM:a,b;                                       |          |            |
|                                           | a                                                | V-160HD  | CAMERA1-16 |
|                                           |                                                  | V-80HD   | CAMERA1-8  |
|                                           |                                                  | VR-120HD | CAMERA1-12 |
|                                           |                                                  | VR-6HD   | CAMERA1-6  |
|                                           |                                                  | V-1-4K   | CAMERA1-5  |
| b                                         | WIDE_FAST, WIDE_SLOW, STOP, TELE_SLOW, TELE_FAST |          |            |
| Response command                          | ACK;                                             |          |            |

## Basic control commands

| CAMZMR      Reset the Zoom for the PTZ camera |                  |          |            |
|-----------------------------------------------|------------------|----------|------------|
| Transmitted command                           | <b>CAMZMR:a;</b> |          |            |
|                                               | <b>a</b>         | V-160HD  | CAMERA1-16 |
|                                               |                  | V-80HD   | CAMERA1-8  |
|                                               |                  | VR-120HD | CAMERA1-12 |
|                                               |                  | VR-6HD   | CAMERA1-6  |
|                                               |                  | V-1-4K   | CAMERA1-5  |
| Response command                              | <b>ACK;</b>      |          |            |

| CAMFC               |            |                 |            | Operate the Focus for the PTZ camera |  |  |  |
|---------------------|------------|-----------------|------------|--------------------------------------|--|--|--|
| Transmitted command | CAMFC:a,b; |                 |            |                                      |  |  |  |
|                     | a          | V-160HD         | CAMERA1-16 |                                      |  |  |  |
|                     |            | V-80HD          | CAMERA1-8  |                                      |  |  |  |
|                     |            | VR-120HD        | CAMERA1-12 |                                      |  |  |  |
|                     |            | VR-6HD          | CAMERA1-6  |                                      |  |  |  |
|                     |            | V-1-4K          | CAMERA1-5  |                                      |  |  |  |
|                     | b          | NEAR, STOP, FAR |            |                                      |  |  |  |
| Response command    | ACK;       |                 |            |                                      |  |  |  |

| CAMAFC                                   |             |          |            |
|------------------------------------------|-------------|----------|------------|
| Set auto focus on/off for the PTZ camera |             |          |            |
| Transmitted command                      | CAMAFC:a,b; |          |            |
|                                          | a           | V-160HD  | CAMERA1-16 |
|                                          |             | V-80HD   | CAMERA1-8  |
|                                          |             | VR-120HD | CAMERA1-12 |
|                                          |             | VR-6HD   | CAMERA1-6  |
|                                          |             | V-1-4K   | CAMERA1-5  |
|                                          | b           | OFF, ON  |            |
| Response command                         | ACK;        |          |            |

| QCAMAFC      Get auto focus status for the PTZ camera |                      |          |            |
|-------------------------------------------------------|----------------------|----------|------------|
| Transmitted command                                   | <b>QCAMAFC:a;</b>    |          |            |
|                                                       | <b>a</b>             | V-160HD  | CAMERA1-16 |
|                                                       |                      | V-80HD   | CAMERA1-8  |
|                                                       |                      | VR-120HD | CAMERA1-12 |
|                                                       |                      | VR-6HD   | CAMERA1-6  |
|                                                       |                      | V-1-4K   | CAMERA1-5  |
| Response command                                      | <b>CAFC:a,b;ACK;</b> |          |            |
|                                                       | <b>a</b>             | V-160HD  | CAMERA1-16 |
|                                                       |                      | V-80HD   | CAMERA1-8  |
|                                                       |                      | VR-120HD | CAMERA1-12 |
|                                                       |                      | VR-6HD   | CAMERA1-6  |
|                                                       |                      | V-1-4K   | CAMERA1-5  |
|                                                       | <b>b</b>             | OFF, ON  |            |



| CAMAEP Turn auto-exposure on/off for the PTZ camera |             |          |            |
|-----------------------------------------------------|-------------|----------|------------|
| Transmitted command                                 | CAMAEP:a,b; |          |            |
|                                                     | a           | V-160HD  | CAMERA1-16 |
|                                                     |             | V-80HD   | CAMERA1-8  |
|                                                     |             | VR-120HD | CAMERA1-12 |
|                                                     |             | VR-6HD   | CAMERA1-6  |
|                                                     |             | V-1-4K   | CAMERA1-5  |
|                                                     | b           | OFF, ON  |            |
| Response command                                    | ACK;        |          |            |

| QCAMAEP Get auto exposure status for the PTZ camera |                      |          |            |
|-----------------------------------------------------|----------------------|----------|------------|
| Transmitted command                                 | <b>QCAMAEP:a;</b>    |          |            |
|                                                     | <b>a</b>             | V-160HD  | CAMERA1-16 |
|                                                     |                      | V-80HD   | CAMERA1-8  |
|                                                     |                      | VR-120HD | CAMERA1-12 |
|                                                     |                      | VR-6HD   | CAMERA1-6  |
|                                                     |                      | V-1-4K   | CAMERA1-5  |
| Response command                                    | <b>CAEP:a,b;ACK;</b> |          |            |
|                                                     | <b>a</b>             | V-160HD  | CAMERA1-16 |
|                                                     |                      | V-80HD   | CAMERA1-8  |
|                                                     |                      | VR-120HD | CAMERA1-12 |
|                                                     |                      | VR-6HD   | CAMERA1-6  |
|                                                     |                      | V-1-4K   | CAMERA1-5  |
|                                                     | <b>b</b>             |          |            |
|                                                     | OFF, ON              |          |            |

| CAMPRExecute Preset Recall for the PTZ camera |            |             |            |
|-----------------------------------------------|------------|-------------|------------|
| Transmitted command                           | CAMPR:a,b; |             |            |
|                                               | a          | V-160HD     | CAMERA1-16 |
|                                               |            | V-80HD      | CAMERA1-8  |
|                                               |            | VR-120HD    | CAMERA1-12 |
|                                               |            | VR-6HD      | CAMERA1-6  |
|                                               |            | V-1-4K      | CAMERA1-5  |
|                                               | b          | PRESET1-100 |            |
| Response command                              | ACK;       |             |            |

| QCAMPR Get current preset number of the PTZ camera |                     |          |            |
|----------------------------------------------------|---------------------|----------|------------|
| Transmitted command                                | <b>QCAMPR:a;</b>    |          |            |
|                                                    | <b>a</b>            | V-160HD  | CAMERA1-16 |
|                                                    |                     | V-80HD   | CAMERA1-8  |
|                                                    |                     | VR-120HD | CAMERA1-12 |
|                                                    |                     | VR-6HD   | CAMERA1-6  |
|                                                    |                     | V-1-4K   | CAMERA1-5  |
| Response command                                   | <b>CPR:a,b;ACK;</b> |          |            |
|                                                    | <b>a</b>            | V-160HD  | CAMERA1-16 |
|                                                    |                     | V-80HD   | CAMERA1-8  |
|                                                    |                     | VR-120HD | CAMERA1-12 |
|                                                    |                     | VR-6HD   | CAMERA1-6  |
|                                                    |                     | V-1-4K   | CAMERA1-5  |
|                                                    | <b>b</b>            |          |            |
|                                                    | N/A, PRESET1-100    |          |            |

## Macro, Sequencer, Graphics Presenter

|                                                                                       |                                          |
|---------------------------------------------------------------------------------------|------------------------------------------|
| MCREX      Execute macro (V-160HD/V-80HD/VR-120HD/VR-6HD)                             |                                          |
| Transmitted command                                                                   | <b>MCREX:a;</b><br><b>a</b> MACRO1–100   |
| Response command                                                                      | <b>ACK;</b>                              |
| QMCNST      Check whether a macro is being executed (V-160HD/V-80HD/VR-120HD/VR-6HD)  |                                          |
| Transmitted command                                                                   | <b>QMCNST:a;</b><br><b>a</b> MACRO1–100  |
| Response command                                                                      | <b>QMCNST:a,b;ACK;</b>                   |
|                                                                                       | <b>a</b> MACRO1–100<br><b>b</b> OFF, ON  |
| SEQSW      Set SEQUENCER ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD)                      |                                          |
| Transmitted command                                                                   | <b>SEQSW:a;</b><br><b>a</b> OFF, ON      |
| Response command                                                                      | <b>ACK;</b>                              |
| QSEQSW      Get status of SEQUENCER (V-160HD/V-80HD/VR-120HD/VR-6HD)                  |                                          |
| Transmitted command                                                                   | <b>QSEQSW;</b>                           |
| Response command                                                                      | <b>QSEQSW:a;ACK;</b><br><b>a</b> OFF, ON |
| SEQAS      Set SEQUENCER Auto Sequence ON/OFF (V-160HD/V-80HD/VR-120HD/VR-6HD)        |                                          |
| Transmitted command                                                                   | <b>SEQAS:a;</b><br><b>a</b> OFF, ON      |
| Response command                                                                      | <b>ACK;</b>                              |
| QSEQAS      Get status of SEQUENCER Auto Sequence (V-160HD/V-80HD/VR-120HD/VR-6HD)    |                                          |
| Transmitted command                                                                   | <b>QSEQAS;</b>                           |
| Response command                                                                      | <b>QSEQAS:a;ACK;</b><br><b>a</b> OFF, ON |
| SEQPV      Set sequence number of SEQUENCER back one (V-160HD/V-80HD/VR-120HD/VR-6HD) |                                          |
| Transmitted command                                                                   | <b>SEQPV;</b>                            |
| Response command                                                                      | <b>ACK;</b>                              |

|                                                                                                                           |                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>SEQPV</b> Set sequence number of SEQUENCER to the previous number or to the beginning (V-160HD/V-80HD/VR-120HD/VR-6HD) |                                                                                            |
| Transmitted command                                                                                                       | <b>SEQPV:a;</b><br><b>a</b> 0–1 0: set sequence back one; 1: set sequence to the beginning |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>SEQNX</b> Advance sequence number of SEQUENCER by one (V-160HD/V-80HD/VR-120HD/VR-6HD)                                 |                                                                                            |
| Transmitted command                                                                                                       | <b>SEQNX;</b>                                                                              |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>SEQJP</b> Set sequence number of SEQUENCER (V-160HD/V-80HD/VR-120HD/VR-6HD)                                            |                                                                                            |
| Transmitted command                                                                                                       | <b>SEQJP:a;</b><br><b>a</b> START, SEQ1–1000                                               |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>GPNC</b> Graphics Presenter: select next contents                                                                      |                                                                                            |
| Transmitted command                                                                                                       | <b>GPNC;</b>                                                                               |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>GPSC</b> Graphics Presenter: select contents (1–124)                                                                   |                                                                                            |
| Transmitted command                                                                                                       | <b>GPSC:a;</b><br><b>a</b> CONTENT1–124                                                    |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>GPHF</b> Graphics Presenter: hide front contents                                                                       |                                                                                            |
| Transmitted command                                                                                                       | <b>GPHF;</b>                                                                               |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>GPHB</b> Graphics Presenter: hide background contents                                                                  |                                                                                            |
| Transmitted command                                                                                                       | <b>GPHB;</b>                                                                               |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |
| <b>GPOA</b> Graphics Presenter: toggle ON AIR                                                                             |                                                                                            |
| Transmitted command                                                                                                       | <b>GPOA;</b>                                                                               |
| Response command                                                                                                          | <b>ACK;</b>                                                                                |

## SYSTEM

|                     |             |
|---------------------|-------------|
| ACS                 | Return ACK  |
| Transmitted command | <b>ACS;</b> |
| Response command    | <b>ACK;</b> |

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| VER                 | Get model name and version number                             |
| Transmitted command | <b>VER;</b>                                                   |
| Response command    | <b>VER:a,b;ACK;</b>                                           |
|                     | <b>a</b> V-160HD, V-80HD, VR-120HD, VR-6HD, V-1-4K Model name |
|                     | <b>b</b> 1.00– Version number                                 |

## Examples of settings

|                     |                              |                                   |
|---------------------|------------------------------|-----------------------------------|
| Transmitted command | <b>VER;</b>                  | Get model name and version number |
| Response command    | <b>VER:V-160HD,2.00;ACK;</b> | V-160HD, Ver.2.00                 |

|                     |                             |
|---------------------|-----------------------------|
| QBSY                | Get busy status of the unit |
| Transmitted command | <b>QBSY;</b>                |
| Response command    | <b>BSY:a;</b>               |
|                     | <b>a</b> OFF, ON            |

|                     |                  |
|---------------------|------------------|
| HDCP                | Set HDCP on/off  |
| Transmitted command | <b>HDCP:a;</b>   |
|                     | <b>a</b> OFF, ON |
| Response command    | <b>ACK;</b>      |

|                     |                   |
|---------------------|-------------------|
| QHDCP               | Get HDCP status   |
| Transmitted command | <b>QHDCP;</b>     |
| Response command    | <b>HCP:a;ACK;</b> |
|                     | <b>a</b> OFF, ON  |

|                     |                                                                          |
|---------------------|--------------------------------------------------------------------------|
| TPT                 | Set Test Pattern                                                         |
| Transmitted command | <b>TPT:a;</b>                                                            |
|                     | <b>a</b> V-160HD                                                         |
|                     | V-80HD OFF, COLORBAR75, COLORBAR100, RAMP, STEP, HATCH, DIAMOND, CIRCLE, |
|                     | VR-120HD COLORBAR75-SP, COLORBAR100-SP, RAMP-SP, STEP-SP, HATCH-SP       |
|                     | VR-6HD                                                                   |
|                     | V-1-4K OFF, COLORBAR                                                     |
| Response command    | <b>ACK;</b>                                                              |

| QTPT Get status of Test Pattern |                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Transmitted command             | <b>QTPT;</b>                                                                                                                       |
| Response command                | <b>TPT:a;ACK;</b>                                                                                                                  |
|                                 | <b>a</b> V-160HD                                                                                                                   |
|                                 | V-80HD OFF, COLORBAR75, COLORBAR100, RAMP, STEP, HATCH, DIAMOND, CIRCLE, COLORBAR75-SP, COLORBAR100-SP, RAMP-SP, STEP-SP, HATCH-SP |
|                                 | VR-120HD                                                                                                                           |
|                                 | VR-6HD                                                                                                                             |
|                                 | V-1-4K OFF, COLORBAR                                                                                                               |

| TTN Set Test Tone Level |                             |
|-------------------------|-----------------------------|
| Transmitted command     | <b>TTN:a;</b>               |
|                         | <b>a</b> OFF, -20, -10, 0dB |
| Response command        | <b>ACK;</b>                 |

| TTN Set Test Tone Level |                                              |
|-------------------------|----------------------------------------------|
| Transmitted command     | <b>TTN:a,b,c;</b>                            |
|                         | <b>a</b> OFF, -20, -10, 0dB Test Tone Level  |
|                         | <b>b</b> 500, 1k, 2kHz Test Tone Frequency L |
|                         | <b>c</b> 500, 1k, 2kHz Test Tone Frequency R |
| Response command        | <b>ACK;</b>                                  |

| QTTN Get Test Tone Level |                                              |
|--------------------------|----------------------------------------------|
| Transmitted command      | <b>QTTN;</b>                                 |
| Response command         | <b>TTN:a,b,c;ACK;</b>                        |
|                          | <b>a</b> OFF, -20, -10, 0dB Test Tone Level  |
|                          | <b>b</b> 500, 1k, 2kHz Test Tone Frequency L |
|                          | <b>c</b> 500, 1k, 2kHz Test Tone Frequency R |

### Commands voluntarily sent from the products in question

| Item              | Transmitted command | Response command | Parameter                                                                                                                                                                                                                 |
|-------------------|---------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Command reception | –                   | <b>ACK;</b>      | –                                                                                                                                                                                                                         |
| Error detection   | –                   | <b>ERR:a;</b>    | a: 0 (syntax error) The received command contains an error.<br>4 (invalid) This has no effect because it is controlled by another setting.<br>5 (out of range error) An argument of the received command is out of range. |